
PULSTAR REACTOR THEORY MANUAL

——
NORTH CAROLINA STATE UNIVERSITY
DEPARTMENT OF NUCLEAR ENGINEERING
NUCLEAR REACTOR PROGRAM



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1 Atomic and Nuclear Physics

1.1 Neutron Reaction

The atom can be considered as being made up of three primary subatomic particles: protons, neutrons and electrons. Protons are particles which possess one unit of positive electrical charge. Electrons are particles which possess one unit of negative electrical charge. Neutrons are particles which possess no electrical charge (neutral). Protons and neutrons are relatively heavy particles and have approximately the same mass. The electron is a relatively light particle and has very little mass.

Neutrons can undergo two types of reactions when interacting with a nucleus:

1. Scattering
2. Absorption

Scattering reactions are divided into two types of reactions.

1. Elastic scattering is when a neutron strikes a nucleus, which is almost always in its ground state, the neutron reappears, and the nucleus is left in its ground state. The total initial kinetic energy (KE) is equal to the total final kinetic energy after the reaction is completed.

$$KE_{n\text{Initial}} + KE_{\text{NucleusInitial}} = KE_{n\text{Final}} + KE_{\text{NucleusFinal}} \quad (1.1)$$

2. Inelastic scattering is when the final KE does not equal the initial KE of the incoming neutron. This occurs when some of the KE of the neutron is lost due to exciting the nucleus.

$$KE_{n\text{Initial}} = KE_{n\text{Final}} + KE_{\text{NucleusExcitation}} + KE_{\text{Nucleus}} \quad (1.2)$$

When a neutron is absorbed, either it can be absorbed and cause fission or it can be absorbed and be captured. A *capture reaction* is when a neutron is absorbed and causes a reaction other than fission such as exciting the nucleus. A *fission reaction* is when the neutron is absorbed and causes the nucleus to split apart resulting in the release of energy.

The extent which nuclear reactions occur are expressed as *cross sections*. A cross section can be thought of as a measure of the probability that a certain reaction will take place. Cross sections exist for many different types of reactions; fission, elastic scattering, and inelastic scattering are examples. Cross sections are a function of neutron energy.

Absorption cross sections are divided into three different regions: 1/v, Resonance, and Fast-Neutron.

1.1.1 1/v Region

In the 1/v region, or low energy region, as neutron kinetic energy increases, the absorption cross section varies inversely with the change in KE. Since KE is energy in motion, the cross section is said to vary inversely with the velocity of the neutron.

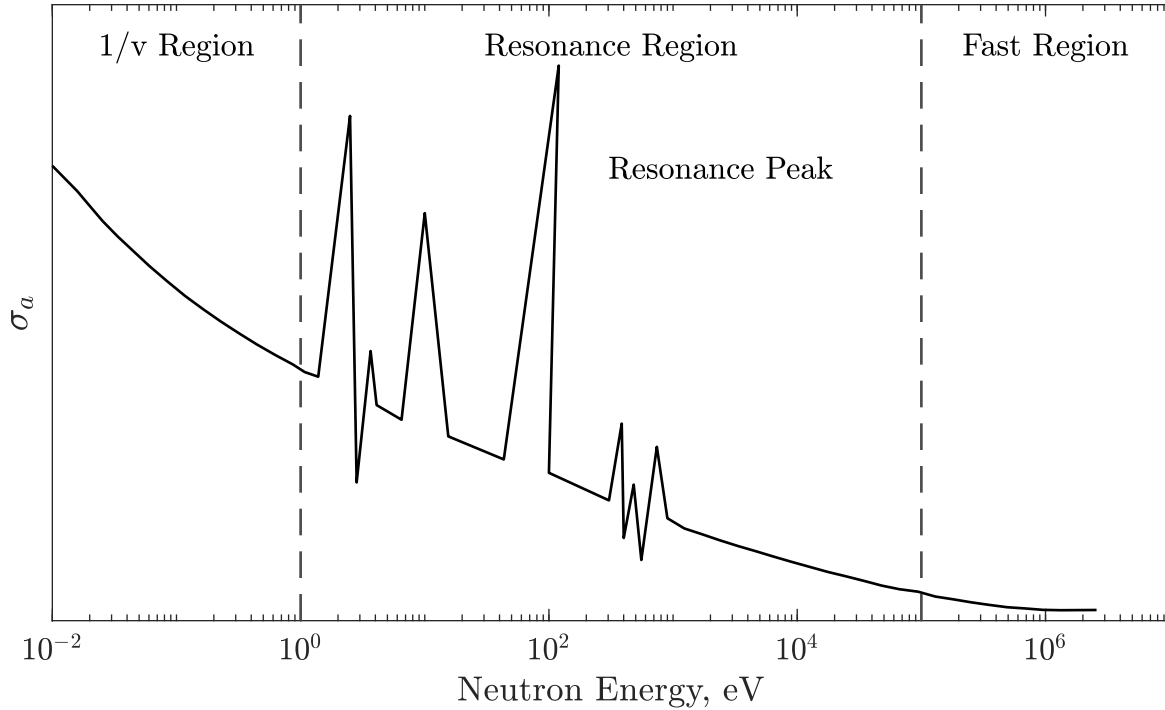


Figure 1.1: Cross section of Uranium.

1.1.2 Resonance Region

The resonance region covers energy levels from approximately 1.0 eV to 1000 eV. The cross section varies with the neutron's energy level. It should be noted that there are large peaks that occur at discrete energy levels. The large peaks are called *resonance peaks*.

1.1.3 Fast Neutron Region

In the fast neutron region, the cross section decreases steadily as the neutron energy increases. From Figure 1.1 it can be noted that as a neutron slows down the probability that a neutron will undergo a reaction increases. Therefore, it is necessary to slow a neutron down (give up energy) so that the neutron can be absorbed and cause fission. A very good material for slowing down (or moderating) a neutron is water. Water slows neutrons down quickly and has a low probability for absorption. There are other materials that are better suited than water for slowing down neutrons, but cost and ease of handling must be taken into account. The PULSTAR uses water as its moderator.

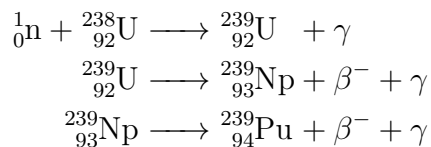
1.2 Fertile, Fissile, and Fissionable Materials

1.2.1 Fertile Nuclide

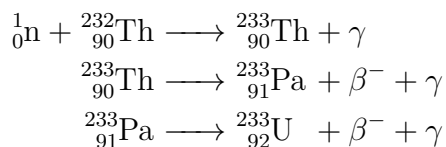
^{232}Th and ^{238}U require fast neutrons to undergo fission, but with neutrons of lower energy values ($< 1 \text{ MeV}$), the nucleus undergo radiative capture. The resultant nuclides, ^{239}Pu and ^{233}U , are fissile nuclides. For this reason, ^{232}Th and ^{238}U are known as fertile nuclides. A

fertile nuclide is a nuclide capable of being transformed, directly or indirectly, into a fissile nuclide by neutron capture. A radiative capture reaction is one by which a nucleus captures a neutron and then immediately emits a gamma particle.

Capture of a neutron by ^{238}U :



Capture of a neutron by ^{232}Th :



1.2.2 Fissile Nuclide

A *fissile nuclide* is a nuclide that will fission by neutrons of all energy levels. There are three fissile nuclide and they are:



In nature, uranium occurs in several different forms, ^{238}U , ^{235}U , and ^{234}U . ^{235}U is the only naturally occurring material that is fissile; ^{233}U and ^{239}Pu are produced from the capture reactions described above.

1.2.3 Fissionable Nuclide

Nuclei such as ^{238}U that do not fission unless struck by high energy neutrons are called fissionable nuclides.

1.3 The Fission Process

In the fission reaction, a nucleus absorbs a neutron of appropriate energy and the resulting compound nucleus breaks up into two parts with the liberation of a considerable amount of energy. At the same time, two or three neutrons are usually emitted, mostly, although not entirely, at the instant of fission. Instantaneous gamma radiation also accompanies the fission process and beta particles, gamma rays, and neutrons are produced over a period of time as the radioactive fission products decay.

Nuclear fission can be compared to a drop of liquid that is set into oscillation and stretches to the point that it splits apart into several different parts. Comparing the nucleus of an atom to the above analogy, when a neutron is absorbed, an excited compound nucleus is formed, or this is to say that the nucleus is oscillating like a liquid drop. The *excitation*

energy (E_{ex}) is equal to the sum of the *kinetic energy* (KE) of the neutron and the neutron *binding energy* (BE). If E_{ex} is greater than the energy required to fission (*critical energy* (E_c)), the nucleus will undergo oscillations and fission. Binding energy is the energy that is required to separate an atom into its individual parts.

$$E_{\text{ex}} = \text{KE} + \text{BE} \geq E_c \rightarrow \text{fission occurs} \quad (1.3)$$

1.3.1 Energy Released from Fission

Table 1.1: Break down of energy released by fission.

	MeV Per Fission
<u>Immediate</u>	
Kinetic energy of fission fragments	168
Kinetic energy of prompt neutrons	5
Instantaneous gamma-ray energy	5
Capture gammas	~ 7
<u>Delayed</u>	
β particles from fission products	7
Neutrons from β decay	~ 10
Gamma rays from fission products	8
Kinetic energy of delayed neutrons	~ 0.04
<u>Total</u>	210

1.4 Effective Multiplication Factor (K_{eff}) and Neutron Life Cycle

1.4.1 Neutron Multiplication

The material which contains nuclides that fission in a reactor system is called the nuclear fuel and it contains more than just fissile nuclei. Freshly fabricated PULSTAR fuel is uranium dioxide (UO_2), enriched in ^{235}U to approximately 4.0%. Naturally occurring uranium consists of about 0.7% ^{235}U and 99.3% ^{238}U . Since the ^{235}U is the isotope of uranium which will fission following the absorption of thermal neutrons, its concentration is increased by the enrichment process.

The production of neutrons from fission leads to the possibility of a *chain reaction*. On the average, 2.43 neutrons are produced from fission. If all of those neutrons caused fission, the fission rate would increase by a factor of 2.43 each generation. That is, if the fission process started with 100 neutrons, 243 neutrons would be produced from those 100 fissions. If all of those 243 neutrons caused fission, it would produce 590 neutrons. The rate at which the

fission rate increases would be determined by the time required for the neutrons produced from fission to be absorbed and to cause another fission. The time for this process is the *generation time*.

Not all of the neutrons produced from fission will cause fission. Some neutrons may leak out of the core without interacting or may be absorbed in material in the core that does not lead to fission. If exactly one of the neutrons produced from each fission causes another fission, the chain reaction is *self-sustaining*. A reactor is *critical* when the fission chain reaction is self-sustaining and stable.

Since the power of the reactor is controlled by controlling the neutron flux in the core, it is useful to examine the various processes which can influence the neutron population. This is referred to as the neutron life cycle, which starts with neutrons produced from fission, and follows those neutrons until more are produced from fission to begin the next generation.

Neutrons produced from fission are “born” at high energies, typically in the vicinity of 1.98 MeV. Even though the probability is low, some of the fast neutrons may be absorbed by both fissile and fissionable nuclide in the fuel and cause *fast fission*. The *fast fission factor*, ϵ , is the ratio of fast neutrons from both fast and thermal fission to the number of fast neutrons from thermal fission alone.

Some of the fast neutrons, which travel at velocities in the thousands of meters per second, may reach the edge of the core and leak out before they have a chance to slow down. The *fast non-leakage probability*, $\mathcal{L}_f$, is the ratio of fast neutrons that start to slow down (or don’t leak from the core) to the number of fast neutrons from fast and thermal fission.

During the *thermalization*, or slowing down process, neutrons exist at energies where resonance capture is likely. Resonance capture results in removal of neutrons from the fission process. The primary isotopes responsible for resonance capture of the intermediate energy neutrons are ^{233}U and ^{240}Pu in the fuel, and silver, indium, and cadmium in the control rods. The *resonance escape probability*, p , is the ratio of the number of neutrons which reach thermal energy, to the number of neutrons which start to slow down.

Neutrons that reach thermal energy either leak out of the core or are absorbed by some material in the core. The ratio of the number of thermal neutrons absorbed in the core to the number of neutrons that reach thermal energy is the *thermal non-leakage probability*, $\mathcal{L}_{\text{th}}$.

All of the remaining thermal neutrons at this point are absorbed in some material in the core, either fuel or non-fuel material. The factor which describes how efficiently those thermal neutrons are utilized by the reactor is the *thermal utilization factor*, f , which is defined as the ratio of thermal neutrons absorbed by the fuel to the number of thermal neutrons absorbed in the core.

The *reproduction factor*, η , is the ratio of the number of fast neutrons from thermal fission to the number of thermal neutrons absorbed in the fuel. There are two reactions that can

take place once thermal neutrons are absorbed by the fuel. The first is capture, a non-fission event, and the second is fission. About five out of six neutrons absorbed by ^{235}U cause fission. One out of six results in a capture reaction, which produces ^{236}U . Each fission produces one or more free neutrons, with the average being 2.43 neutrons per fission of ^{235}U . Other isotopes have different ratios of the capture to fission cross sections, and different numbers of free neutrons per fission.

The combined effect of each of the above six factors, is expressed by the *effective multiplication factor*, K_{eff} . It is defined as the ratio of fast neutrons from thermal fission in one generation to the fast neutrons from thermal fission in the previous generation. It provides information concerning the rate of change of neutron population in the core. A K_{eff} of 1 means that the neutron population in the core is not changing from generation to generation, or the reactor is exactly critical. A K_{eff} greater than unity means the reactor is *supercritical*, or neutron population and therefore power production is increasing. A *subcritical* reactor is one in which K_{eff} is less than unity, with neutron population and power both decreasing. Refer to Figure 1.2 for a pictorial representation of the life cycle of a neutron in a nuclear reactor. The formula (six factor formula) for computing K_{eff} is:

$$K_{\text{eff}} = \epsilon f \eta p \mathcal{L}_f \mathcal{L}_{\text{th}} \quad (1.4)$$

Reactivity, ρ , is defined as the fractional change in neutron population per generation. It is defined mathematically as:

$$\rho = \frac{K_{\text{eff}} - 1}{K_{\text{eff}}} \quad (1.5)$$

A critical reactor has a reactivity of zero. A reactor with positive reactivity is supercritical, and a reactor with negative reactivity is subcritical. The net reactivity is directly related to the rate of change of reactor power.

Reactivity may be expressed in several different units including:

- $\frac{\Delta K}{K}$
- percent $\frac{\Delta K}{K}$ ($\% \frac{\Delta K}{K}$)
- percent millirho (pcm) or $10^{-5} \frac{\Delta K}{K}$

Example 1.1 :

$$0.00001 \frac{\Delta K}{K} = 0.001\% \frac{\Delta K}{K} = 1 \text{ pcm}$$

Reactivity is very useful because it can be used to compare the effects on the fission process due to various changes in plant parameters. For example, a one degree increase in coolant temperature on the PULSTAR when it is critical would cause the reactor to be subcritical by a reactivity of 3.9 pcm. A withdrawal of control rods by an amount of 3.9 pcm on the PULSTAR may make the reactor supercritical by that amount. It is observed that the two reactivity effects are of equal magnitude but opposite directions, which means that the two effects in this case would cancel each other out and return the reactor to a critical condition.

As will be discussed in detail, in later sections, many plant parameters affect the values of K_{eff} and reactivity in the core, such as:

1. moderator temperature
2. reactor power level
3. reactor power history
4. control rod position
5. fuel burn-up

1.5 Subcritical Multiplication

A critical reactor is one in which the fission process alone produces a self-sustaining chain reaction, that is, the effective multiplication factor is unity. A subcritical reactor is one in which the fission process alone does not sustain a chain reaction, but rather would result in power decreasing exponentially for an indefinite period of time if it were not for the presence of neutrons from a non-fission source. Neutrons in the core, produced by processes other than neutron-induced fission, are referred to as source neutrons. These source neutrons are multiplied in the fuel by the fission process and allow a constant neutron population even though the effective multiplication factor is less than one. This process, referred to as subcritical multiplication, is the means by which the operator controls the reactor when shutdown and during startup to the point of achieving criticality.

1.5.1 Source Neutrons

When the reactor is shutdown for some time, the neutron population can decay to a level below the useful range of the source range nuclear instrumentation. It is necessary to provide a source of neutrons to make up for the loss of fission neutrons so that a visible, or reliable, indication of neutron population exists. The source neutrons are multiplied by the fission process and produce a constant neutron population, with $K_{\text{eff}} < 1$. If the source is of sufficient strength, the neutron population attained by subcritical multiplication will be in the visible range of indication. Thus, the Reactor Operator is provided with a reliable, real-time knowledge of the behavior of the neutron population and the effective multiplication factor, K_{eff} . With this information, the operator can avoid inadvertent criticality during fuel loading, and criticality can be approached in a controlled manner during a reactor startup. Source neutrons come from sources that are either intrinsic or installed. Both types of sources are described below.

Intrinsic sources are present in the reactor because they are produced as a result of interactions between, or reactions within, the materials used to build the reactor. The most important intrinsic sources are mentioned below.

1. Photoneutron reactions with deuterium are the most significant intrinsic source immediately following shutdown from power operation. An inventory of fission products must first be established, because deuterium requires high-energy gamma radiation for the reaction that produces a neutron $[{}^2_1\text{H}(\gamma, n){}^1_1\text{H}]$.

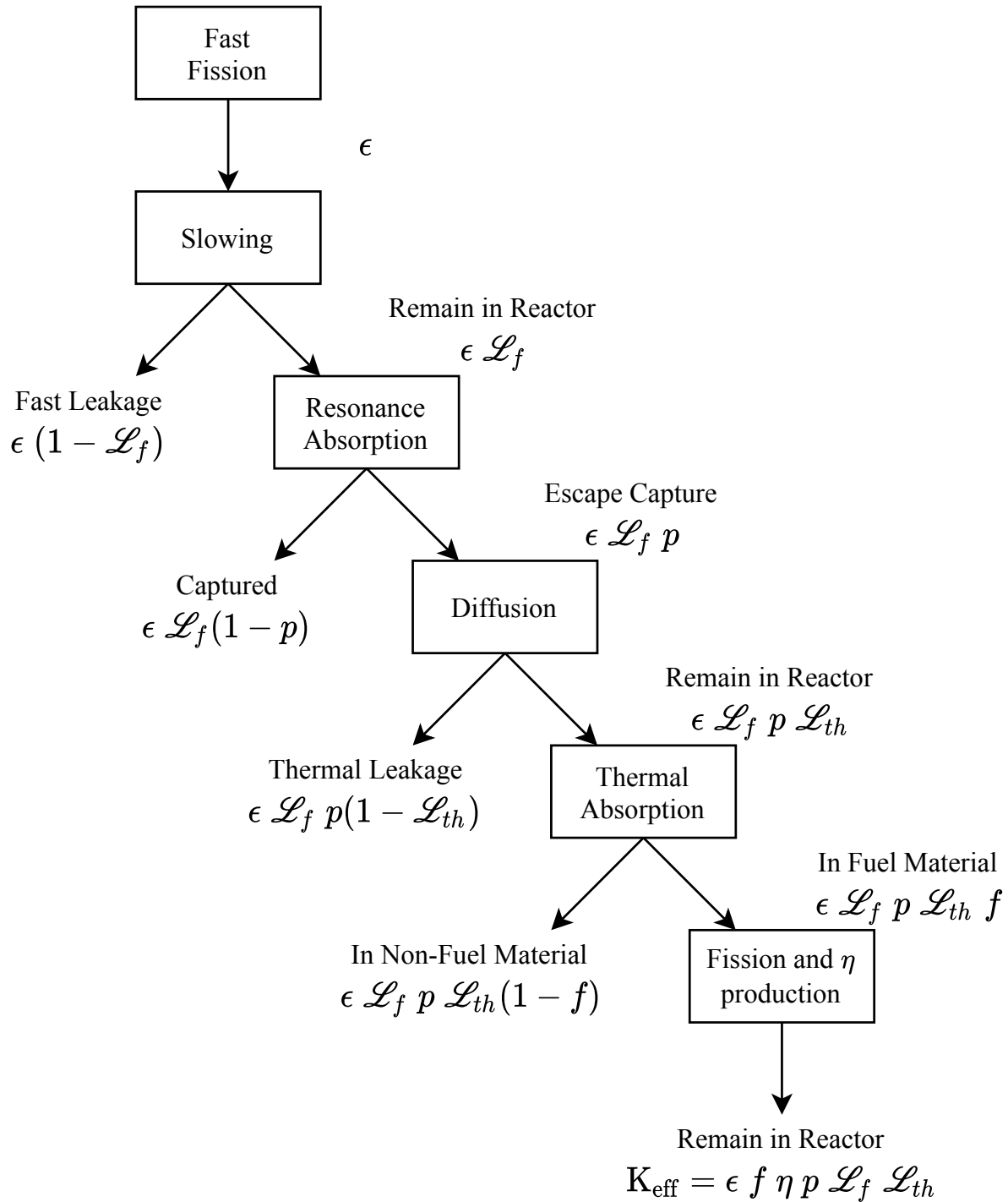


Figure 1.2: Neutron life cycle.

2. Spontaneous fission occurs in ^{235}U , ^{238}U and some trans-uranic elements, especially ^{242}Cm and ^{244}Cm . This is most significant late in core life.
3. The (α, n) reaction with the ^{18}O in UO_2 fuel molecules is another source. The α particles for this reaction come from decay of ^{235}U , ^{238}U , ^{239}Pu and other transuranic elements.
4. Photofission reactions with ^{238}U and transuranic elements also produce neutrons, but their contribution is small when compared to that of the three sources previously mentioned.

Intrinsic sources may not provide enough neutrons to produce a sizable neutron population in a new core or after a long shutdown. Installed sources increase the shutdown neutron population. The PULSTAR utilizes a Pu-Be source in which the α particle, for the (α, n) reaction with ^9Be , comes from the decay of Pu. The neutrons that are produced are then multiplied by the fuel.

1.5.2 Neutron Multiplication

Note: The derivations of the equations in this section are not required knowledge, but are provided so that a complete understanding of the different neutron processes can be achieved.

Consider a subcritical reactor, with $K_{\text{eff}} < 1$. N neutrons will be multiplied by the fuel to produce $K_{\text{eff}} \times N$ fission neutrons. Since $K_{\text{eff}} < 1$, $K_{\text{eff}} \times N$ is less than N , or one might say that $N - (K_{\text{eff}} \times N)$ neutrons were lost from the fission chain. If S represents the total contribution of source neutrons to the process, then the number of neutrons from both fission and the source is $(K_{\text{eff}} \times N) + S$, which is the number to begin the next generation. Now, if the number of neutrons lost from the fission chain, $N - (K_{\text{eff}} \times N)$ or $N(1 - K_{\text{eff}})$, is exactly equal to the source neutrons added to the process, there will be no net change in neutron population from generation to generation.

At steady state, the number of neutrons lost from the fission chain is exactly equal to the source neutrons added:

$$\begin{aligned} N - NK_{\text{eff}} &= S \\ N(1 - K_{\text{eff}}) &= S \end{aligned} \tag{1.6}$$

The ratio of the number of neutrons from the source and fuel to the number of neutrons from the source is called the subcritical multiplication factor, M :

$$M = \frac{S + N_f}{S} \tag{1.7}$$

where: S = number of source neutrons and N_f = number of fission neutrons.

Since the number of neutrons from the source and fuel is the total neutron population, N , Equations 1.6 and 1.7 can be combined as follows:

$$M = \frac{N}{S} = \frac{N}{(1 - K_{\text{eff}})} = \frac{1}{1 - K_{\text{eff}}} \quad (1.8)$$

Alternatively:

$$N = \frac{S}{1 - K_{\text{eff}}} \quad (1.9)$$

Now consider starting the multiplication process with no neutrons in the core. The non-fission neutron sources will produce a quantity of neutrons equal to S which will be multiplied in the fuel. The first generation will produce:

$$N_f = SK_{\text{eff}} \text{ fission neutrons.} \quad (1.10)$$

Succeeding generations will multiply these neutrons and another quantity of source neutrons as follows:

Example 1.2 :

$$\begin{aligned} N_f &= SK_{\text{eff}} + SK_{\text{eff}}^2 \text{ after 2 generations} \\ N_f &= SK_{\text{eff}} + SK_{\text{eff}}^2 + SK_{\text{eff}}^3 \text{ after 3 generations} \end{aligned}$$

$$N_f = SK_{\text{eff}} + SK_{\text{eff}}^2 + SK_{\text{eff}}^3 \dots SK_{\text{eff}}^n \text{ after } n \text{ generations} \quad (1.11)$$

Substituting the n^{th} generation value of N_f into Equation 1.7 yields:

$$M = \frac{S + SK_{\text{eff}} + SK_{\text{eff}}^2 + SK_{\text{eff}}^3 \dots SK_{\text{eff}}^n}{S} \quad (1.12)$$

$$M = \frac{S(1 + K_{\text{eff}} + K_{\text{eff}}^2 + K_{\text{eff}}^3 \dots K_{\text{eff}}^n)}{S} \quad (1.13)$$

$$M = 1 + K_{\text{eff}} + K_{\text{eff}}^2 + K_{\text{eff}}^3 \dots K_{\text{eff}}^n \quad (1.14)$$

$$M = \sum_{n=0}^n K_{\text{eff}}^n \quad (1.15)$$

Equation 1.15 is a geometric series that can be simplified by the following steps. First, multiply both sides of the equation by $(1 - K_{\text{eff}})$:

$$M(1 - K_{\text{eff}}) = (1 - K_{\text{eff}}) \sum_{n=0}^n K_{\text{eff}}^n \quad (1.16)$$

The right-hand side of the equation simplifies to $(1 - K_{\text{eff}}^{n+1})$. (Try this multiplication yourself with $n = 3$). Finally:

$$\begin{aligned} M(1 - K_{\text{eff}}) &= (1 - K_{\text{eff}}^{n+1}) \\ M &= \frac{(1 - K_{\text{eff}}^{n+1})}{(1 - K_{\text{eff}})} \end{aligned} \quad (1.17)$$

Note that this is similar to the results obtained in Equation 1.8 for M at equilibrium. The difference is the transient term, $(1 - K_{\text{eff}}^{n+1})$. Equation 1.17 can be used to find the neutron population after any number of generations, n , by multiplying the initial population (assumed to be equal to the source strength, S) by the multiplication, M :

$$\begin{aligned} N_n &= S(M_n) \\ N_n &= S \left[\frac{(1 - K_{\text{eff}}^{n+1})}{(1 - K_{\text{eff}})} \right] \end{aligned} \quad (1.18)$$

Since the quantity of interest is generally not the neutron population after n generations but the equilibrium neutron population, Equation 1.18 needs to be solved for equilibrium as in Equation 1.17. After many generations, $(1 - K_{\text{eff}}^{n+1})$ approaches 0 (for $K_{\text{eff}} < 1$). For instance, if $K_{\text{eff}} = 0.92$, $(1 - K_{\text{eff}}^{n+1}) = 9.7 \times 10^{-7}$ after $n = 165$ generations. We can then say that $(1 - K_{\text{eff}}^{n+1})$ is approximately 0 at equilibrium and simplify Equation 1.18 for neutron population at equilibrium, N_{eq} .

$$N_{\text{eq}} = \frac{S}{1 - K_{\text{eff}}} \quad (1.19)$$

Now, since the indicated count rate is proportional to the neutron population, the relationship that describes the change in count rate for a change in K_{eff} can be developed. For simplicity, K_{eff} will be indicated as K_{eff} with a subscript to associate a value of K_{eff} with a corresponding value of equilibrium count rate. The source strength, S , is assumed to be constant.

$$\begin{aligned} \text{CR} &\propto \frac{S}{1 - K_{\text{eff}}} \\ \text{If: } \text{CR}_1 &\propto \frac{S}{1 - K_{\text{eff},1}} \quad \text{CR}_2 \propto \frac{S}{1 - K_{\text{eff},2}} \\ \text{Then: } \text{CR}_1(1 - K_{\text{eff},1}) &\propto S \propto \text{CR}_2(1 - K_{\text{eff},2}) \end{aligned} \quad (1.20)$$

$$\text{Therefore } \text{CR}_2 = \text{CR}_1 \frac{1 - K_{\text{eff},1}}{1 - K_{\text{eff},2}} \quad (1.21)$$

We can now use Equation 1.21 to solve for equilibrium count rate following a change in K_{eff} from $K_{\text{eff},1}$ to $K_{\text{eff},2}$. This is demonstrated in the following example:

Example 1.3:

If $\text{CR}_1 = 10$ cps at $K_{\text{eff},1} = 0.95$, what is CR_2 at $K_{\text{eff},2} = 0.97$?

$$\text{CR}_2 = \text{CR}_1 \left(\frac{1 - K_{\text{eff},1}}{1 - K_{\text{eff},2}} \right) = 10 \left(\frac{1 - 0.95}{1 - 0.97} \right) = 10 \left(\frac{0.05}{0.03} \right) = 16.7 \text{ cps}$$

1.5.3 Subcritical Multiplication Characteristics

Now that the equations associated with a subcritical reactor have been presented, the associated reactor characteristics can be explored. Table 1.2 shows the neutron population

in a hypothetical reactor for different values of K_{eff} and an assumed neutron source of 200 neutrons per generation. The count rates for each value of K_{eff} are plotted against neutron generations in Figure 1.3.

By examining this information, three characteristics of subcritical multiplication can be identified and explained.

The first characteristic of subcritical multiplication is that equal changes in K_{eff} or reactivity do not result in equal changes in equilibrium neutron population. Comparing the increase in equilibrium population from $K_{\text{eff}} = 0.7$ to 0.8 and from 0.8 to 0.9, it can be seen that progressively larger changes occur for the same increase in K_{eff} .

Table 1.2: Neutron population v.s. K_{eff} .

		K_{eff}				
		0.6	0.7	0.8	0.9	0.95
Neutron Population After n Generations	$n = 0$	200	200	200	200	200
	$n = 1$	320	340	360	380	390
	$n = 2$	392	438	488	542	571
	$n = 3$	435	507	590	688	742
	$n = 4$	461	555	672	819	905
	$n = 5$	477	588	738	937	1060
	$n = 6$	486	612	790	1043	1207
	$n = 7$	492	628	832	1139	1346
	$n = 8$	495	640	866	1225	1479
	$n = 9$	497	648	893	1303	1605
	$n = 10$	498	653	914	1372	1725
Equilibrium Population		500	667	1000	2000	4000
% of Equil. in 10 Gen.		99.6	97.9	91.4	68.6	43.1
Increase in Equil. Pop.		N/A	167	333	1000	2000
Generations to Equil.		26	38	61	130	268

The second characteristic is that as K_{eff} approaches 1.0, more generations are required to reach equilibrium; therefore, more time is required to reach equilibrium after each change in K_{eff} . When Equation 1.19 was simplified to determine equilibrium count rate, K_{eff} was assumed to be 0 at equilibrium. A large number of generations is required to reach equilibrium. The neutron population is considered to be at equilibrium when the increase in count rate is insignificant.

Example 1.4:

Consider the case where $K_{\text{eff}} = 0.9$ and $S = 100$ neutrons/generation. The equilibrium population will be:

$$CR_{eq} = \frac{S}{1 - K_{\text{eff}}} = \frac{100}{1 - 0.9} = 1000 \text{ cps}$$

After 20 generations:

$$CR_{20} = S \left(\frac{1 - K_{\text{eff}}^{n+1}}{1 - K_{\text{eff}}} \right) = 100 \left(\frac{1 - 0.9^{20}}{1 - 0.9} \right) = 891 \text{ cps}$$

After 50 generations:

$$\begin{aligned} CR_{50} &= 995 \text{ cps} \\ \Delta CR &= CR_{50} - CR_{20} = 104 \text{ cps} \end{aligned}$$

After 100 generations:

$$\begin{aligned} CR_{100} &= 999.98 \text{ cps} \\ \Delta CR &= CR_{100} - CR_{50} = 4.98 \text{ cps} \end{aligned}$$

After 130 generations, $CR_{130} = 999.99899$ cps. The increase from this value to the equilibrium of 1000 cps is insignificant. As a guideline, when $K_{\text{eff}}^{n+1} = 10^{-6}$ the change in count rate in succeeding generations will be insignificant, even when the neutron population is very large. Using this assumption, Table 1.3 shows that as K_{eff} approaches 1.0 a dramatic increase in the number of generations to equilibrium occurs; hence, more time is required to reach equilibrium.

Table 1.3: Generations to equilibrium.

K_{eff}	Generations to Equilibrium
0.9	130
0.99	1373
0.999	13808
0.9999	138147

The third characteristic of subcritical multiplication provides the Reactor Operator with a method to determine the proximity of the reactor to criticality by observation of the source range recorder trace.

From Equation 1.21:

$$CR_2 = CR_1 \frac{1 - K_{\text{eff},1}}{1 - K_{\text{eff},2}}$$

If $(1 - K_{\text{eff},2})$ is reduced by a factor of 2, then CR_2 will increase by a factor of 2. In other words, if K_{eff} is taken halfway to one from its present value, the count rate will double. This is seen in Table 1.2 as K_{eff} changes from 0.6 to 0.8, 0.8 to 0.9, and again from 0.8 to 0.95. Each time the difference between K_{eff} and 1.0 is cut in half, equilibrium count rate doubles. Count rate doubling gives a good indication of how close the reactor is to criticality. If CR doubles 5 times, the reactor will be very near critical. This is apparent from the fact that

approximately one-half of the reactivity needed to go critical is added each time count rate doubles. After 5 doubles, only 3% (0.5^5), of the initial negative reactivity remains in the core.

The three characteristics of subcritical multiplication are readily observable on a source range recorder trace. Figure 1.4 shows a source range recorder trace for a reactor startup. While subcritical, each incremental rod withdrawal results in a steady increase in count rate that gradually levels off to a new subcritical multiplication level. Once the reactor is critical, however, a steady increase in count rate is observed that does not round-out to a vertical

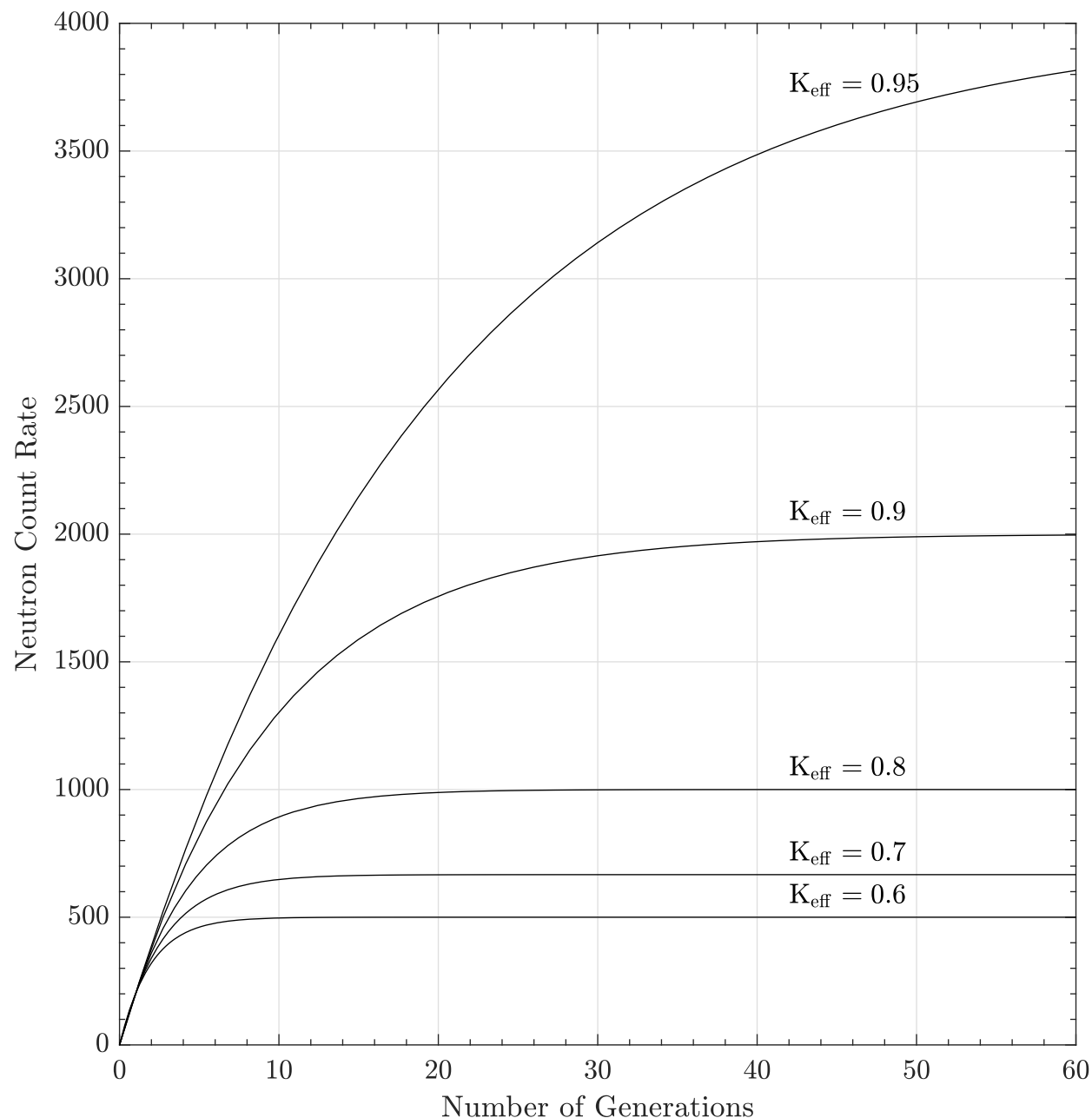


Figure 1.3: Time to equilibrium as a function of K_{eff} .

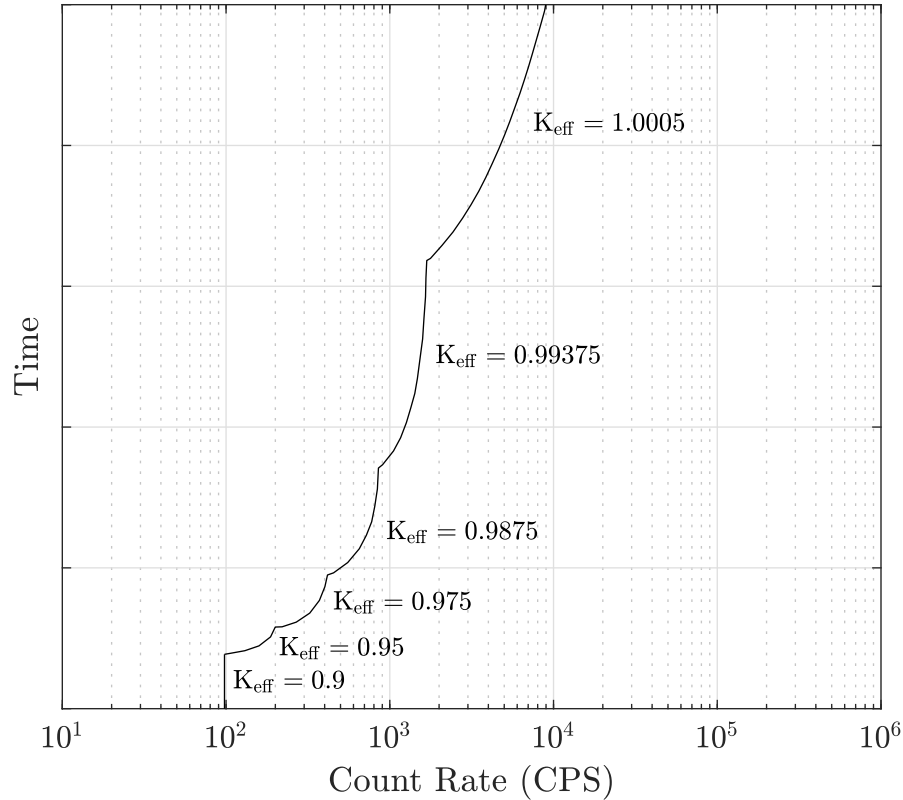


Figure 1.4: Reactor startup traces.

line. A logarithmic chart recorder will display the power rise as a straight line with a positive slope, and a linear chart recorder will display the power rise as a curved line, concave downward, i.e., increasing at an increasing rate. Both of these indications are indicative of an exponential increase in power.

One can also observe the different types of startup rates encountered during a reactor startup. An apparent startup rate is caused by the change in neutron population from one subcritical multiplication level to another. This startup rate returns to 0 when equilibrium count rate is reached. A stable startup rate indicates an exponential increase in reactor power and stays above 0 when rod motion stops. The transition from an apparent startup rate to a stable positive startup rate marks the point of criticality.

In summary, the three most important points with regard to subcritical multiplication are:

1. Equal changes in either K_{eff} or reactivity do not result in equal changes in equilibrium neutron population.
2. As K_{eff} approaches 1.0, more time is required to reach a new equilibrium neutron population.
3. If the reactor is taken halfway to criticality in terms of K_{eff} , count rate will double, and after 5 doubles the reactor will be approximately critical.

1.5.4 1/M Plots

A 1/M plot is very useful for predicting criticality. Recall that the factor by which the initial count rate changes when K_{eff} is changed is the multiplication, M . The final count rate divided by the initial count rate is the multiplication:

$$M = \frac{CR_2}{CR_1} = \left(\frac{1 - K_{\text{eff},1}}{1 - K_{\text{eff},2}} \right) \quad (1.22)$$

As $K_{\text{eff},2}$ approaches 1.0, M approaches infinity. This relationship cannot be used to plot an approach to criticality. However, as M approaches infinity, $1/M$ approaches zero.

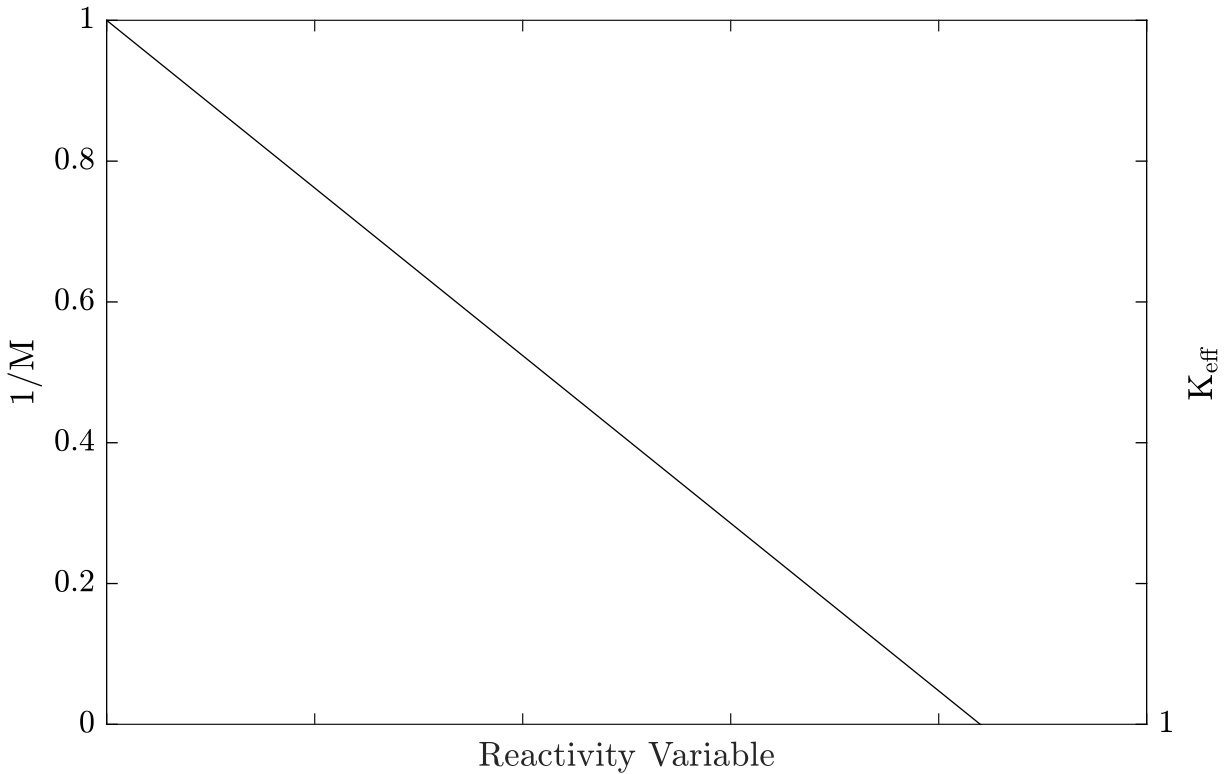


Figure 1.5: Ideal 1/M plot.

This latter relationship can be used to plot an approach to criticality since M , and thus $1/M$, can be related to count rate as shown above:

$$\text{If } M = \frac{CR_2}{CR_1}, \text{ then } 1/M = \frac{CR_1}{CR_2} \quad (1.23)$$

Where: CR_1 is the initial count rate and CR_2 is the final count rate.

As K_{eff} increases from $K_{\text{eff},1}$ to $K_{\text{eff},2}$, the count rate increases and $1/M$ decreases. An ideal 1/M plot is shown in Figure 1.5. The horizontal axis is the reactivity variable. The reactivity is expressed in terms of mass of fuel, number of fuel assemblies added to the core, amount

of rod withdrawal, dilution of boron, time, and temperature, etc..

An actual $1/M$ plot can vary significantly from the ideal case shown in Figure 1.5. The assembly of a reactor core involves a continual change of volume (changes within the six-factor formula). Also, a neutron source is concentrated, that is, a point source, rather than distributed. These factors, combined with the fuel-detector-source geometry, make it unlikely to obtain the ideal curve of Figure 1.5.

Figure 1.6 illustrates one possible departure from linearity for a $1/M$ plot during fuel loading. There is a large neutron density, or flux, at the detector because the neutron source is nearby. The count rate from the detector is, therefore, initially due to the source neutrons and is not very sensitive to fission neutrons until the mass of fuel is near the critical value. This type of $1/M$ plot is called non-conservative because it predicts criticality later than it actually occurs.

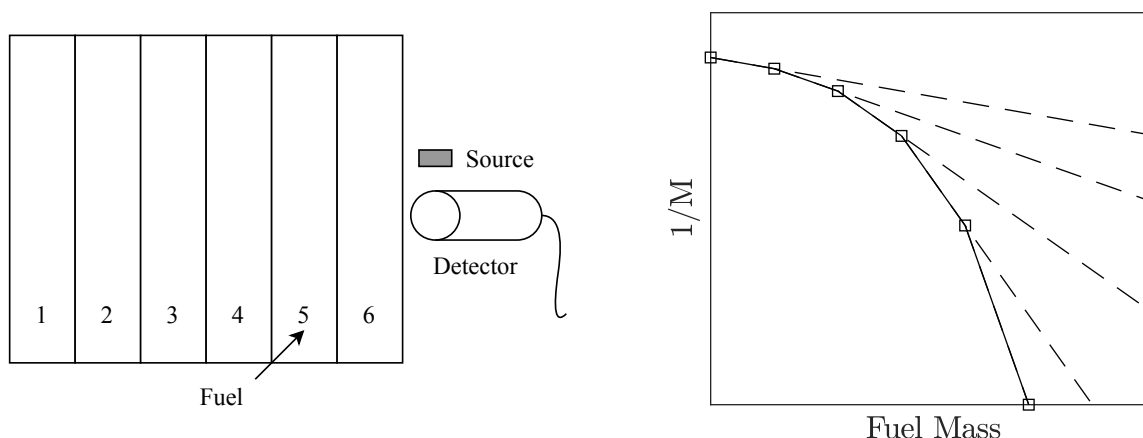


Figure 1.6: Non-conservative $1/M$ plot.

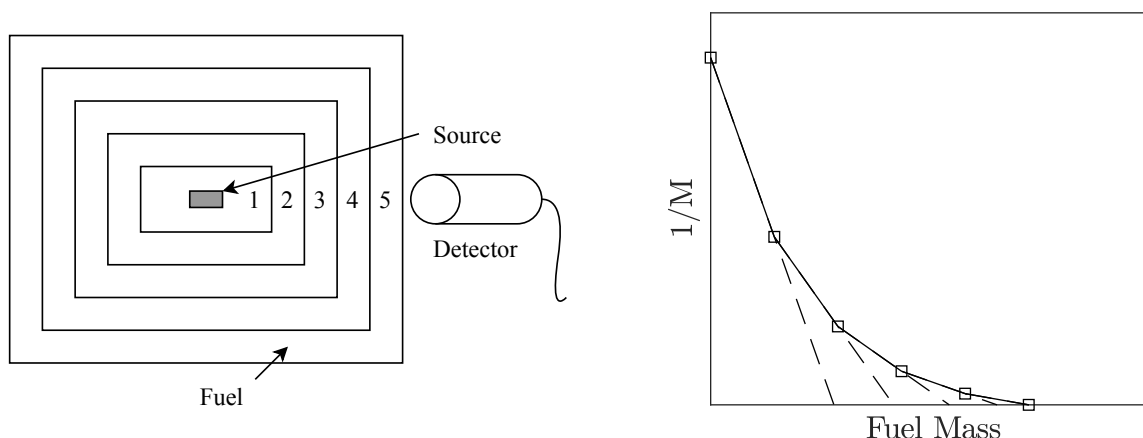


Figure 1.7: Conservative $1/M$ plot.

Now consider a $1/M$ plot with a different arrangement as shown in Figure 1.7. The first fuel addition multiplies the source, and the detector sees a large increase in neutron flux

caused by fission in the fuel. This causes a large initial increase in the count rate and a corresponding drop in $1/M$. A $1/M$ plot with this shape is called a conservative plot because it predicts that the reactor will achieve criticality sooner than it really does.

The existence of these diverse shapes of $1/M$ plots (and all combinations of shapes in between) helps to explain the reasoning behind and importance of using two (or more) independent detector systems during a fuel load.

In general, three factors can lead to a non-ideal $1/M$ plot: geometry, unequal reactivity additions, and control rod shadowing.

1. Geometry is the source-detector-fuel arrangement discussed in the previous paragraph and is best dealt with by the use of multiple detectors. The most conservative $1/M$ curve from the detector responses should be used for subsequent reactivity additions.
2. Unequal reactivity additions cause non-linearity because the reactivity axis of the $1/M$ curve (labeled as number of fuel assemblies, rod position, etc.) assumes that each unit of the independent variable has equal reactivity worth; however, this is not necessarily true. Fuel assemblies loaded in different locations have different worths, and reactivity additions due to rod withdrawal vary with the differential rod worth.
3. Control rod shadowing involves placement of the detector behind a control rod with respect to the source. If a control rod is positioned in the core between the source and detector, the detector will be shielded (or shadowed) by the control rod, which depresses neutron flux in the immediate vicinity. The result is a non-conservative $1/M$ plot since the detector does not see the full effect of the fuel assembly additions behind the control rod.

Since it is not generally possible to accurately determine K_{eff} from multiplication measurements during the early approach to criticality, the information sought is the number of assemblies (inches of rod withdrawal, etc.) required to become critical. One should be prepared for the possibility of $1/M$ plots that deviate significantly from linearity, especially if the fuel-detector-source arrangement is unfavorable.

A $1/M$ plot is developed as follows:

1. Establish the reference count rate, C_0 , after confirming that the detectors are operable. Obtain separate C_0 's for each detector channel. The C_0 's are the starting point for the $1/M$ plot with $C_0/C_0 = 1 = 1/M_0$ (See Figure 1.8).
2. The first reactivity addition should be no more than $1/2$ the amount needed for criticality. After the reactivity addition, allow count rate to stabilize. Failure to do this can cause a non-conservative $1/M$ plot. Obtain new count rates from each detector channel and plot the corresponding values of $1/M$.
3. Using the two $1/M$ points now plotted, draw a line that intersects the reactivity variable axis. This is the predicted critical condition of fuel mass, rod height, etc.

- Subsequent reactivity additions should be less than half of the estimated reactivity to achieve criticality. Figure 1.8 illustrates the method of joining the last two points on the $1/M$ plot and the linear extrapolation to the point where $1/M = 0$ for the new prediction of criticality.

A $1/M$ plot is used during initial core loading of an unknown core configuration to ensure that the reactor will not go critical on fuel assembly addition. The reactor is not expected to go critical with the entire core loaded. However, this is an engineering prediction and must be confirmed by using a $1/M$ plot.

A $1/M$ plot can be used for any startup to provide reliable indication of the proximity to criticality.

1.6 Questions

- Define source neutrons and describe their significance to reactor operations.
- Draw a log SRM channel trace and calculate the equilibrium count rate (if the reactor is subcritical) for the following:

Time	Count Rate
0	SRM reads 100 cps, $K_{\text{eff}} = 0.995$
1	add 200 pcm
2	add 200 pcm
3	add 200 pcm

- Draw a $1/M$ plot for the following:

Fuel Assembly Load	Count Rate
0	10
4	20
6	25
10	34
15	62

- Give two possible reasons for the shape of the plot in Question 3.

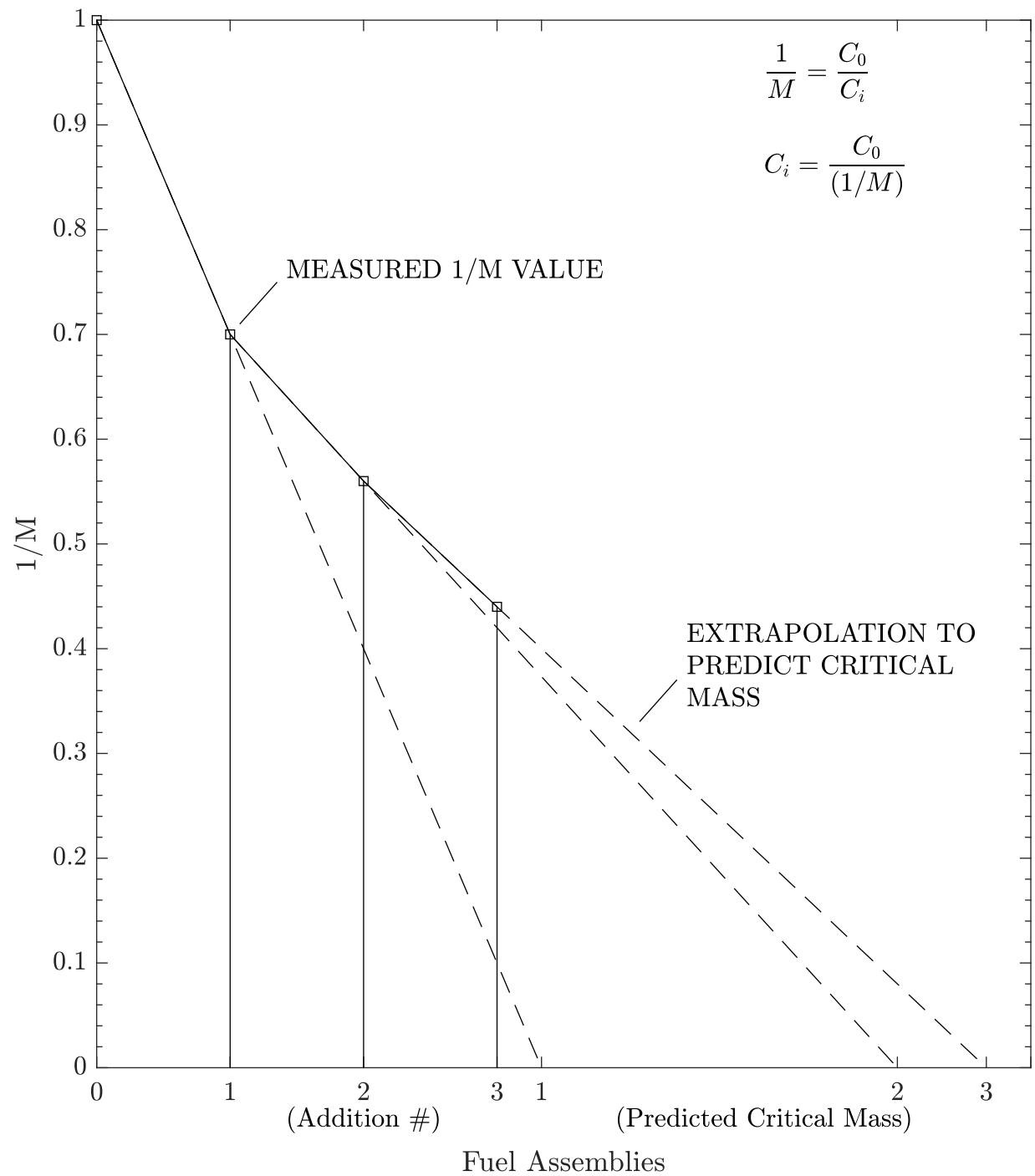


Figure 1.8: Example 1/M plot.

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- [3] American Nuclear Society, *Glossary of Terms in Nuclear Science and Technology*. La Grange, Park, IL, 1986.
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2 Nuclear Parameters

2.1 Reactivity and Startup Rate

Reactor behavior in the intermediate range is not affected by source neutrons, and there are no significant reactivity effects due to fuel or moderator heating while below the point of adding heat. For these reasons, reactor kinetics equations that relate reactivity to the rate of power increase can be better discussed by looking at reactor behavior in the intermediate range. This section will examine reactivity effects on a critical reactor in the intermediate range

2.2 Positive Stable Startup Rate

To understand the response of the reactor to a change in reactivity, one must consider the effect of delayed neutrons and prompt neutrons. Figure 2.1 illustrates the generation of prompt and delayed neutrons. Neutrons that appear (born) immediately after fission are called *prompt neutrons*. The time from fission until a prompt neutron is born is, 1×10^{-14} sec.

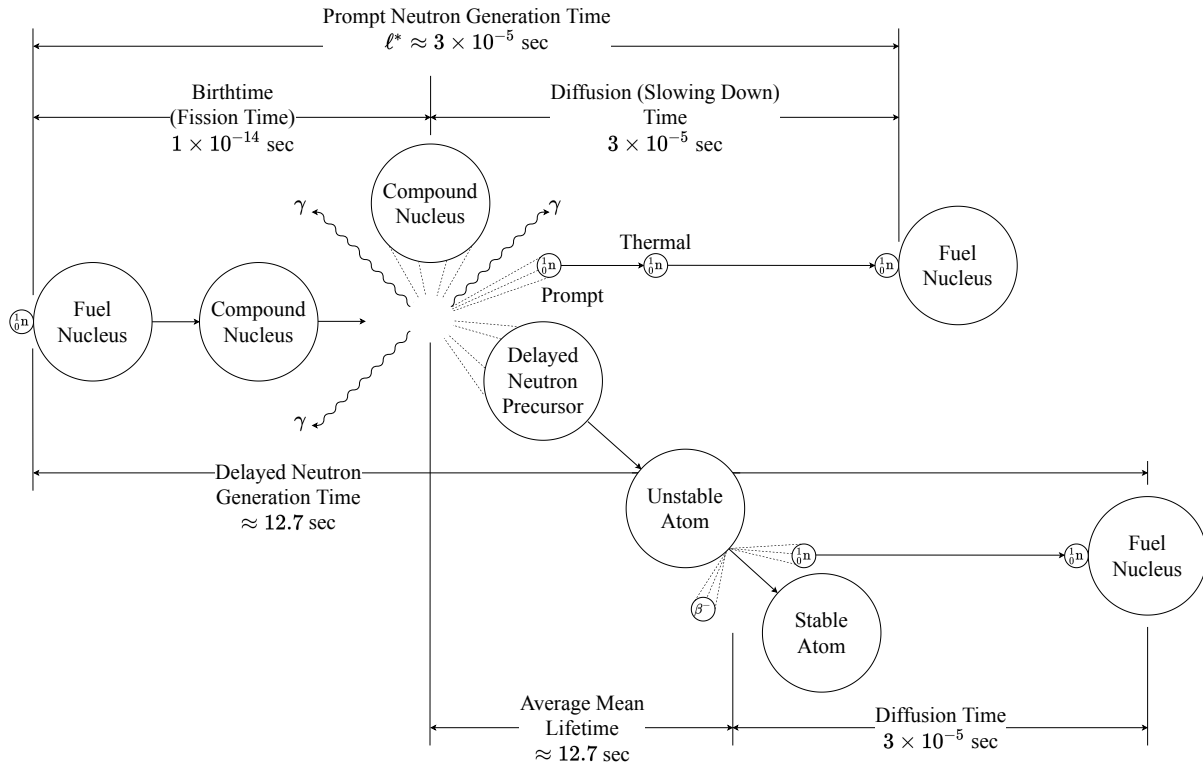


Figure 2.1: Prompt and delayed neutron generation times.

Delayed neutrons are neutrons that are born from the radioactive decay of unstable atoms formed from the fission process. A fraction of the fission neutrons is withheld in each generation and stored away in delayed neutron precursor (DNP or delayed neutron emitters).

These delayed neutron precursors are divided into six groups with $t_{sfrac12}$ ranging from approximately 0.23 seconds to approximately 55.6 seconds. The rate at which they are released is proportional to the number of precursor atoms in each group (C_i) and the decay constant of the group (λ_i) as illustrated in Figure 2.2.

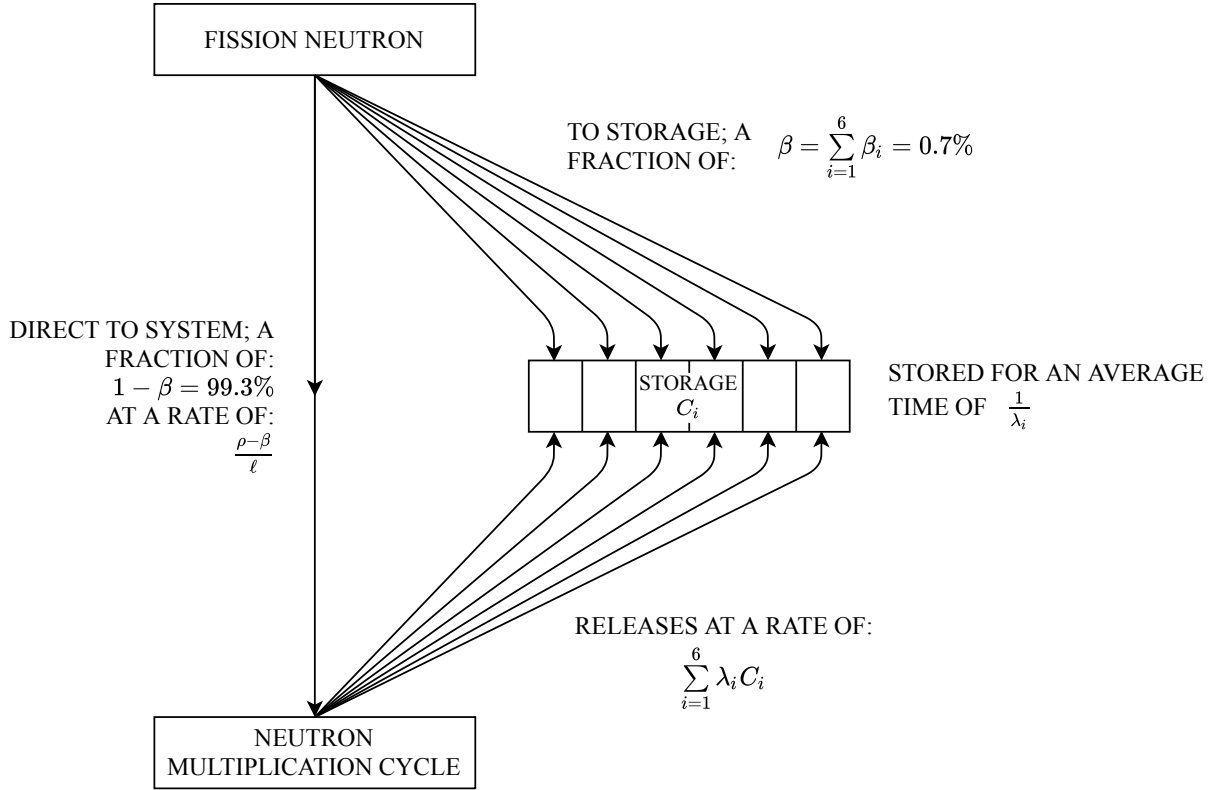


Figure 2.2: Neutron multiplication cycle.

At any instant of time we have the withholding of a portion of the fission neutrons in that generation on one hand and the release of a portion of delayed neutrons from storage on the other hand. When the reactor is at constant power level, there is a balance between these two processes; the rate of release of delayed neutrons equals the rate at which they are stored away.

As the reactivity is increased by withdrawing a control rod, the balance between storage and release of delayed neutrons is upset. Initially, there is a sharp rise in the neutron population due to the multiplication of the prompt fission neutrons. However, as the fission rate increases, so does the rate at which neutrons are stored away in precursors, while the rate at which delayed neutrons are released, which depends on the number of precursor atoms in storage from previous generations, remains effectively the same for a short time. The upset in the balance slows the increase of reactor power, thus providing control. After a short transition time the system adjusts itself to the new conditions, and the neutron population grows exponentially. Figure 2.3 illustrates the response of the reactor to a positive reactivity insertion made on a critical reactor, low in the intermediate range.

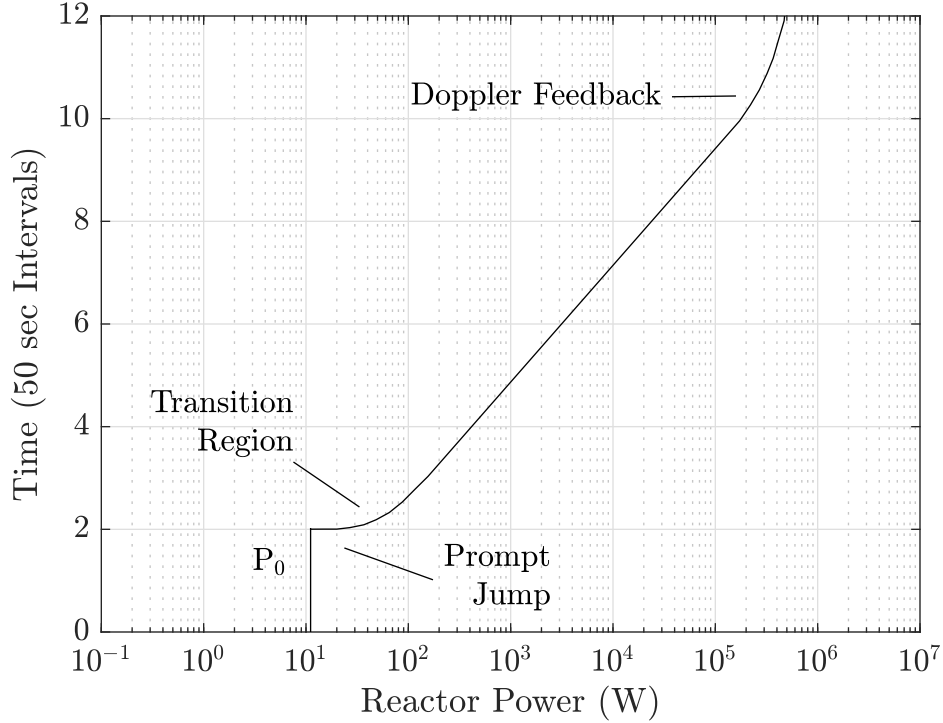


Figure 2.3: Reactor response to positive reactivity.

An exponential growth means the quantity under consideration increases by the same factor in each unit of time. In the case of reactor power, the factor is K_{eff} and the unit of time is the average neutron generation time. *Prompt neutron generation time* (ℓ_p) is the time from the fission event until the prompt neutron is absorbed, and since the time for fission is very short (1×10^{-14} sec) compared to the lifetime of a prompt neutron (3×10^{-5} sec) prompt neutron generation time can be considered to be 3×10^{-5} sec. When only prompt neutrons are considered, the neutron life cycle (or generation time), takes place in a very short time. By looking at what the *effective neutron generation time* ($\bar{\ell}$) would be if all neutrons were born as prompt neutrons and then looking at the affect that delayed neutrons have on the generation time the importance of delayed neutron is apparent.

Looking at the equation:

$$P_f = P_0 e^{\left(\frac{K_{\text{eff}} - 1}{\bar{\ell}}\right)t} \quad (2.1)$$

and considering only prompt neutrons as the generation time, we can see that reactor power increases at a rate which would be uncontrollable by the operator.

Example 2.1 :

Assume:

1. Initial Power (P_0) is 1 watt.
2. Add 9.9 pcm which results in a slightly supercritical reactor ($K_{\text{eff}} = 1.0001$).
3. Time: 2 sec.
4. $\bar{\ell} = 3 \times 10^{-5}$ sec.

$$P_f = 1 \text{ W} \left[e^{\left(\frac{1.0001 - 1}{3 \times 10^{-5} \text{ sec}} \right) (2 \text{ sec})} \right] = 785 \text{ W}$$

The *average precursor mean life* for all delayed-neutrons is approximately 12.7 sec and taking into account that the fission time and the slowing and diffusion time of the delayed neutrons is very short (3×10^{-5} sec) when compared to the overall mean life, the delayed neutron generation (ℓ_d) time can be said to be 12.7 sec. The effective generation time ($\bar{\ell}$) is equal to:

$$\bar{\ell} = \ell_d(\beta) + \ell_p(1 - \beta) \quad (2.2)$$

where: β = delayed neutron fraction.

Example 2.2 :

$$(12.7)(0.0065) = (3 \times 10^{-5})(0.9935) = 0.083 \text{ sec}$$

Substituting the $\bar{\ell}$ into Equation 2.1 yields the following:

$$P_f = 1 \text{ W} \left[e^{\left(\frac{1.0001 - 1}{0.083 \text{ sec}} \right) (2 \text{ sec})} \right] = 1.0024 \text{ W}$$

Therefore if the reactor was critical on only prompt neutrons then power would increase at a rate that would be impossible for the operator to control, but because under normal operations delayed neutrons are needed to sustain the fission process and they increase the average neutron generation time, the reactor can be controlled by the operator.

2.3 β_{eff} and Reactor Response

Delayed neutron fraction, β , is the fraction of fission neutrons that are born delayed and the *effective delayed neutron fraction*, β_{eff} , is the ratio of the mean number of fissions caused by delayed neutrons to the mean total number of fissions caused by delayed plus prompt neutrons, this value for the PULSTAR can be found in Reference 6. Since delayed neutrons are needed to continue the fission process, they are very important to the operator. Delayed neutrons are born at lower energy levels than prompt neutrons therefore they are less likely

to undergo fast leakage and resonance capture, this make them more effective at causing fission than prompt neutrons. As discussed previously delayed neutrons in a reactor increase the effective generation time of the reactor making the fission process easier to control during power changes.

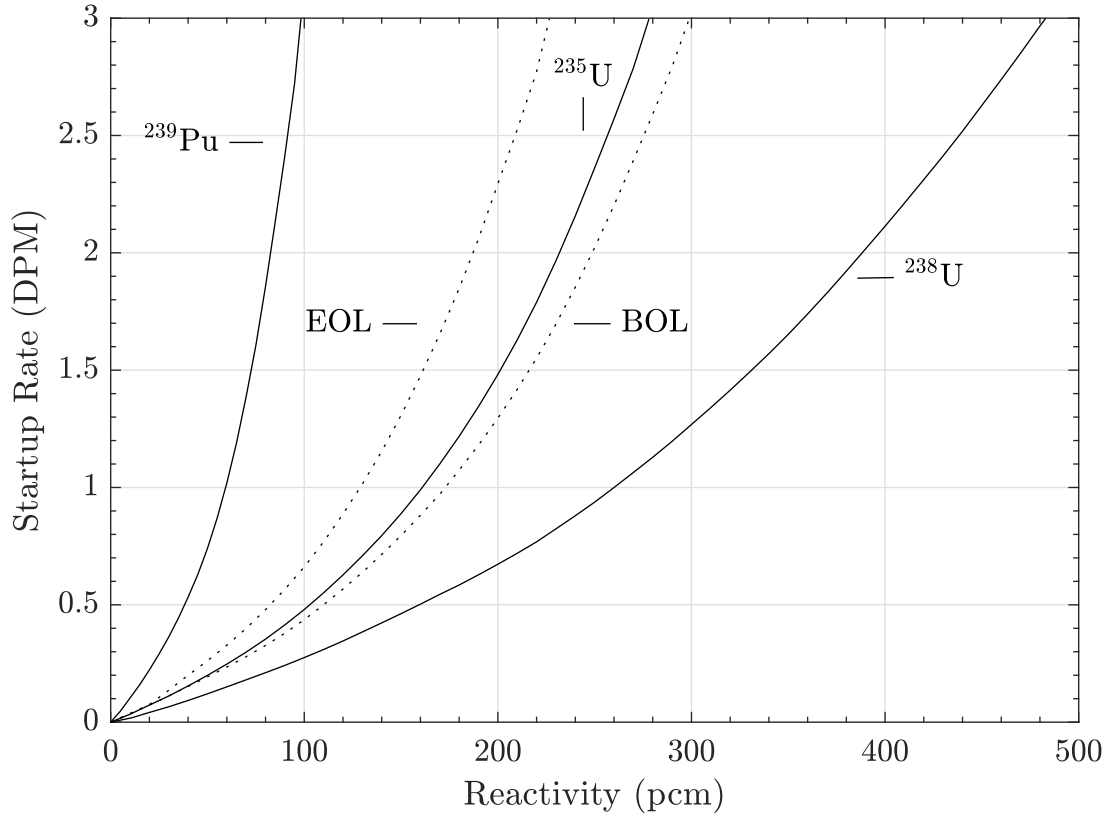


Figure 2.4: Reactor fuel characteristics.

The amount that the effective generation time increases is determined by the effective delayed neutron fraction, β_{eff} and the half-lives of the DNP's. The value of β_{eff} is obtained from the delayed neutron fractions of each fuel nuclide, a weighting factor that accounts for the fraction of power generated by each nuclide, and an importance factor that accounts for the difference in the probability of fission, between prompt and delayed neutrons.

Figure 2.4 shows the startup rates caused by varying amounts of reactivity in a reactor fueled by a single isotope, ^{235}U , ^{238}U , or ^{239}Pu . Figure 2.4 also shows the beginning of life (BOL) and end of life (EOL) response for reactor fuel combinations that produce β_{eff} values of 730 pcm (BOL) and 548 pcm (EOL).

The BOL curve in Figure 2.4 assumes no ^{239}Pu in the core. As the fraction of power produced from ^{239}Pu increases, as it does over core life, the reactor response curve shifts toward the ^{239}Pu response curve. This means that the reactor will respond faster at the end of core life than at the beginning of core life for a given reactivity change.

2.4 Negative Reactivity

On the sudden insertion of negative reactivity, as is the case for a dropped control rod, we observe a prompt drop and then the much slower decay of the delayed neutron precursors as illustrated in Figure 2.5. After enough time, a negative, stable SUR is reached as determined from the Inhour equation. However, for negative reactivity insertions greater than $\frac{1}{2}\beta_{\text{eff}}$ (the effective delayed neutron fraction), the value of the stable SUR is weakly dependent on the reactivity insertion.

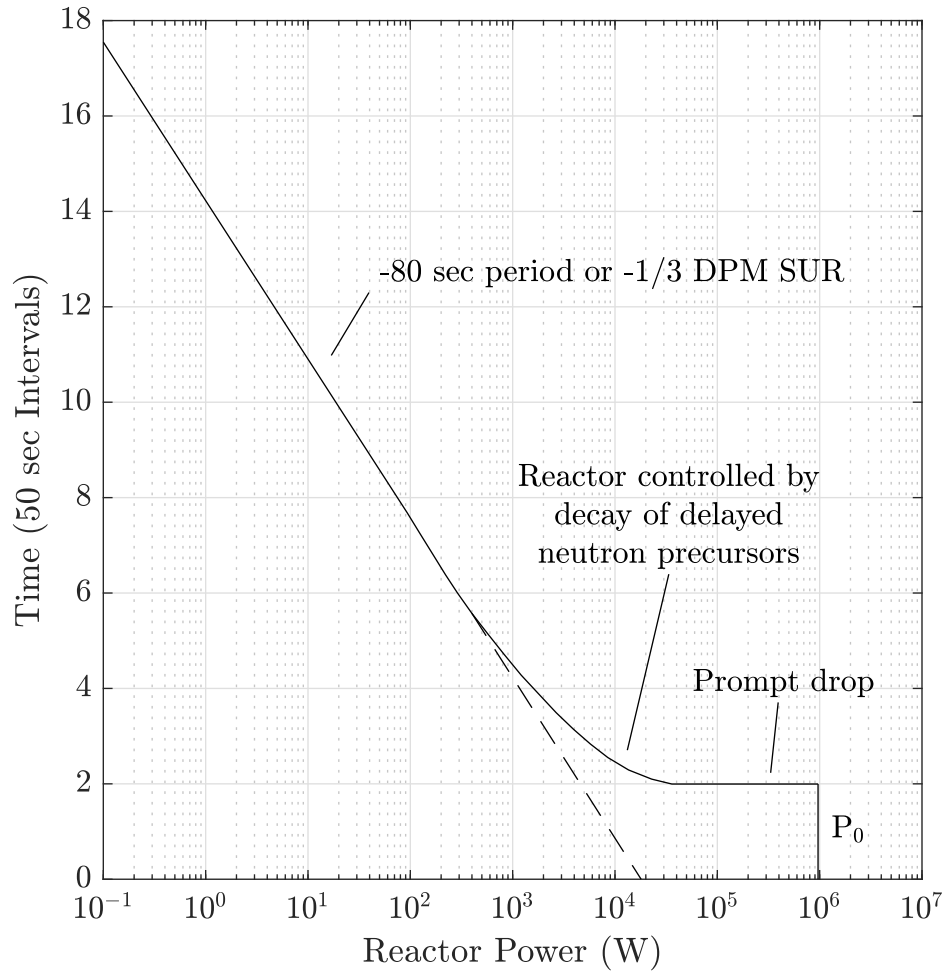


Figure 2.5: Reactor response to a sudden insertion of negative reactivity.

For negative reactivity insertions of large magnitudes, reactor power decreases at the same rate as the activity of the longest-lived group of delayed neutron precursors. Thus, the stable period following a large negative reactivity insertion approaches -80 seconds (or a stable SUR of -1/3 DPM).

2.5 Exponential Power Equations

For simplicity of calculation, factors such as e (2.718), 10 and 2, along with their corresponding units of time, are used to describe a change in reactor power. Examples of exponential power equations are given below:

$$\begin{aligned} P_f &= P_0 e^{t/\tau} & \tau &= \text{period (in seconds)} \\ P_f &= P_0 2^{t/dt} & dt &= \text{doubling time (in seconds)} \\ P_f &= P_0 10^{t/DT} & DT &= \text{decade time (in minutes)} \end{aligned} \quad (2.3)$$

These terms have a physical interpretation, i.e., the period is the time for power to change by a factor of e , the doubling time is the time for power to change by a factor of 2, etc. In practical application the reciprocal of the decade time, called startup rate, is more commonly used.

$$P_f = P_0 10^{\text{SUR } t} \quad \text{SUR} = \text{startup rate (in minutes)} \quad (2.4)$$

It is useful to express the quantities reactor period, SUR, and doubling time in terms of each other. The following equations can be mathematically derived to show commonly used relationships:

$$\begin{aligned} \text{SUR} &= \frac{26.1}{\tau} \\ \tau &= 1.44 dt \\ \text{SUR} &= \frac{18.1}{dt} \end{aligned} \quad (2.5)$$

2.6 Inhour Equation

The stable period, τ , is related to reactivity, ρ , by the Inhour equation, which has seven terms:

$$\rho = \frac{\ell^*}{K\tau} + \frac{\beta_1}{1 + \lambda_1 \tau} + \frac{\beta_2}{1 + \lambda_2 \tau} + \dots + \frac{\beta_6}{1 + \lambda_6 \tau} \quad (2.6)$$

This equation can be written in shorthand form as:

$$\rho = \frac{\ell^*}{K\tau} + \sum_{i=1}^6 \frac{\beta_i}{1 + \lambda_i \tau} \quad (2.7)$$

where the symbol $\sum$ indicates the summation of six similar terms.

The Inhour equation can be used to determine the reactivity necessary to produce a given SUR or the SUR that will result from a given reactivity insertion. This can be accomplished with a graphed solution to the Inhour equation (Figure 2.6) or by using the one-group approximation to the Inhour equation. This approximation is made by replacing the seven terms of the Inhour equation with one term that contains an effective delayed neutron fraction and an effective delayed neutron precursor decay constant as follows:

$$\rho = \frac{\beta_{\text{eff}}}{1 + \lambda_{\text{eff}} \tau} \quad (2.8)$$

where λ_{eff} is the effective delayed neutron precursor decay constant.

The value of λ_{eff} is actually a function of reactivity but can be approximated as $0.1 \frac{1}{\text{s}}$ for slightly supercritical, $0.078 \frac{1}{\text{s}}$ for critical, and $0.012 \frac{1}{\text{s}}$ for shutdown. This equation is useful for small reactivity insertions ($\rho \ll \beta_{\text{eff}}$) which produce slow startup rates, such as those normally experienced at nuclear plants. It should be noted that for large amounts of reactivity ($\rho > \frac{1}{2}\beta_{\text{eff}}$) the one-group approximation will give considerable error as compared to the seven-term graphed solution of the Inhour equation.

The one-group approximation to the Inhour equation may be written in terms of SUR by substituting $\frac{26.1}{\text{SUR}}$ for τ in the preceding equation and solving for SUR:

$$\text{SUR} = \frac{26.1\lambda_{\text{eff}}\rho}{\beta_{\text{eff}} - \rho} \quad (2.9)$$

2.7 Reactivity Coefficients

As conditions change in the reactor the changing parameters have an effect on the reactivity of the core and thus effect reactor power. This section will discuss changing plant parameters,

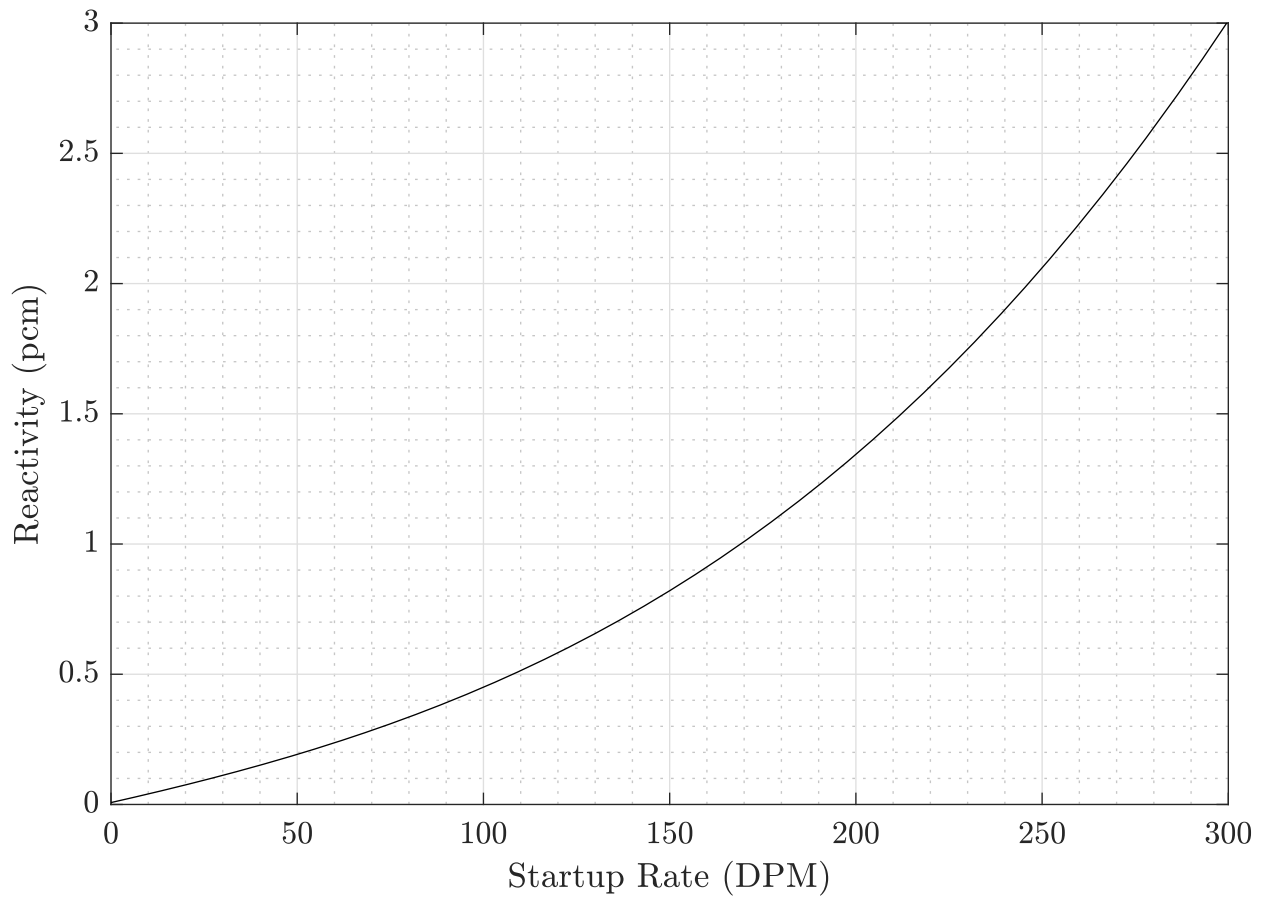


Figure 2.6: Inhour equation graph.

how this affects the different factors in the neutron life cycle and the effect this has on the reactivity of the core.

2.7.1 Moderator Temperature Coefficients

Moderator temperature is one of the process variables that affects reactivity. A negative temperature coefficient produces negative feedback that tends to dampen power transients, which leads to inherent stability of the reactor system. A positive moderator temperature coefficient decreases the stability of the reactor control system, which requires operator or automatic control system action to control reactor transients. The direction and magnitude of this reactivity change is of great importance from the standpoint of reactor safety and control. Therefore, the reactor is designed such that under normal operating conditions, an increase in moderator temperature adds negative reactivity.

The change in reactivity resulting from a one-degree change in moderator temperature is called the moderator temperature coefficient:

$$\alpha_{T,\text{mod}} = \frac{\Delta\rho}{\Delta T_{\text{mod}}} \quad (2.10)$$

As moderator temperature increases, moderator density decreases, as shown in Figure 2.7. This decrease in density decreases the number of moderator atoms in the core, which causes the slowing down length and diffusion length of the neutrons to increase. The effect of the temperature increase is summarized below:

1. *Resonance escape probability* (p) decreases – The longer slowing down length means that neutrons stay in the resonance region for a longer time and have an increased probability of resonance capture. This is a negative reactivity effect.
2. *Thermal utilization factor* (f) increases – The decrease in the number of moderator atoms in the core causes the macroscopic absorption cross section of the moderator to decrease, which increases the thermal utilization factor. This is a positive reactivity effect.
3. *Non-leakage probabilities* ($\mathcal{L}_f$ and $\mathcal{L}_{th}$) decrease – The longer the slowing down length and diffusion length, the greater the probability of a neutron escaping from the core. This increase in leakage is a negative reactivity effect.

The combination of the three phenomena mentioned make the moderator temperature coefficient negative.

One of the parameters that affects the magnitude of the moderator temperature coefficient is moderator temperature. As moderator temperature increases, the change in density per degree change in moderator temperature increases (See Figure 2.7). This causes the change in (f) and the change in (p) to increase proportionally in magnitude. The net effect is that the moderator temperature coefficient becomes more negative at higher moderator temperatures.

A change in moderator temperature can add either positive or negative reactivity depending upon the direction of temperature change based on a reference temperature. The

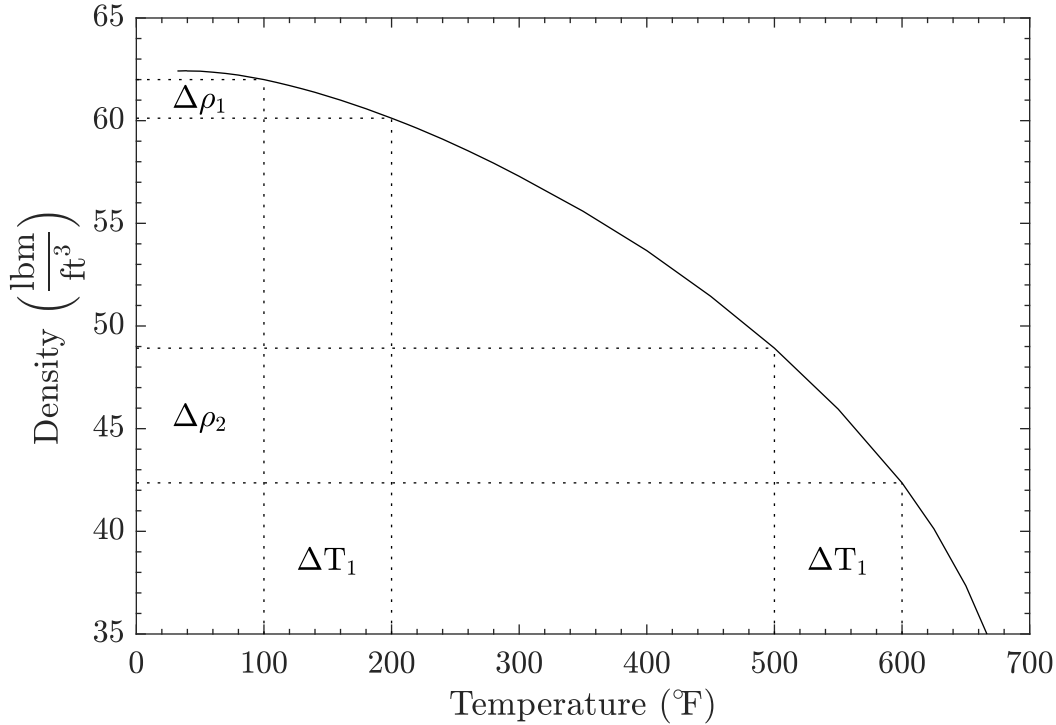


Figure 2.7: Density of water versus temperature.

PULSTAR's the reference temperature is 70 °F. The magnitude of the reactivity effect is dependent upon the moderator temperature coefficient $\alpha_{T,\text{mod}}$. The current value of the moderator temperature coefficient at PULSTAR can be found in Reference 6.

2.7.2 Fuel Temperature Coefficient

In a low enrichment reactor, the temperature of the fuel has a significant effect on the reactivity of the reactor. The change in reactivity resulting from a one-degree change in fuel temperature is called the fuel temperature coefficient or the Doppler coefficient:

$$\alpha_{T,\text{fuel}} = \frac{\Delta\rho}{\Delta T_{\text{fuel}}} \quad (2.11)$$

The low enrichment fuel used in the PULSTAR contains a large fraction of ^{238}U . The capture cross-section spectrum of this isotope contains large resonance peaks where the absorption cross-section for neutrons is several orders of magnitude greater than at neighboring energies. As the temperature of the fuel is increased, these resonance peaks broaden and flatten (Figure 2.8), which is known as Doppler broadening. For an individual atom of ^{238}U , the change in shape of the capture cross-section peak with temperature has no effect on capture probability because the decrease in the cross-section at the midpoint energy of a resonance peak is balanced by the increase on either side, since the area under the curve remains constant. However, the combined effect of Doppler broadening of the resonance peaks of all atoms in a fuel pellet affects the number of neutrons absorbed within the resonance interval due to self-shielding.

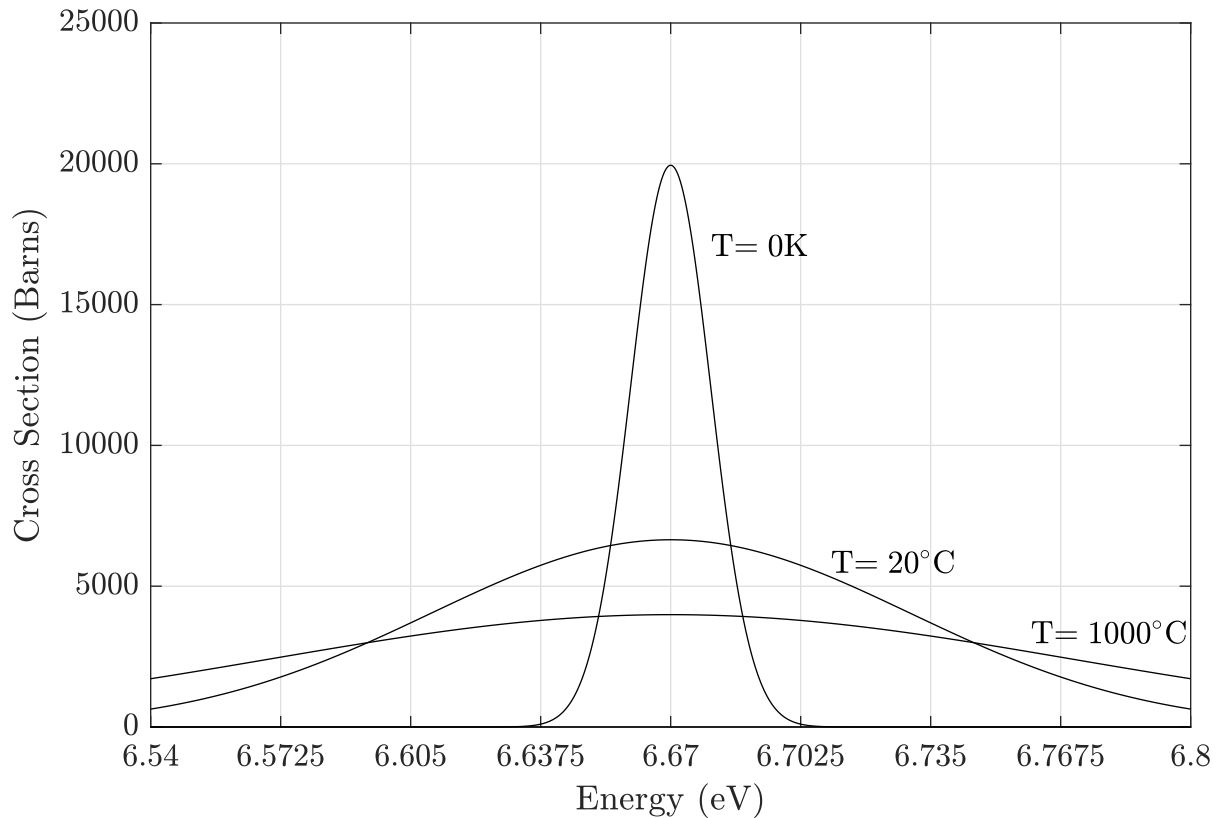


Figure 2.8: Doppler broadening of ^{238}U resonance peaks.

Self-shielding is a phenomenon occurring in heterogeneous cores. The mean free path of a neutron at a given energy is inversely proportional to the capture cross-section at that energy. For neutrons at energies corresponding to resonance peaks, the mean free path is small compared to the dimensions of the fuel pellet. As a result, most of the resonance captures occur in the outer part of the fuel pellet, effectively shielding the inner part of the fuel from resonance-energy neutrons.

The effect of a fuel temperature increase on neutrons of different energies is depicted in Figure 2.9, which shows the change of a single resonance peak in ^{238}U as a result of a temperature increase. Consider the effect of this change on neutrons at energies A and B. Energy A is the centroid energy of the resonance. Energy B is outside the resonance peak at low temperature so the absorption cross-section for neutrons at this energy is very low. Neutrons entering the fuel at energy A will most likely be absorbed within a short distance after entering the fuel, while neutrons at energy B are not absorbed before leaving the fuel region.

The effect of a fuel temperature increase is that the absorption cross-section for neutrons at energy A decreases, as shown in Figure 2.9, while the cross-section for energy B increases. Neutrons at energy A can travel farther in the fuel before being absorbed, but they are still absorbed in the fuel. Note that this does not affect reactivity because neutrons at energy A were absorbed at low temperatures as well. The significant effect on neutron absorption

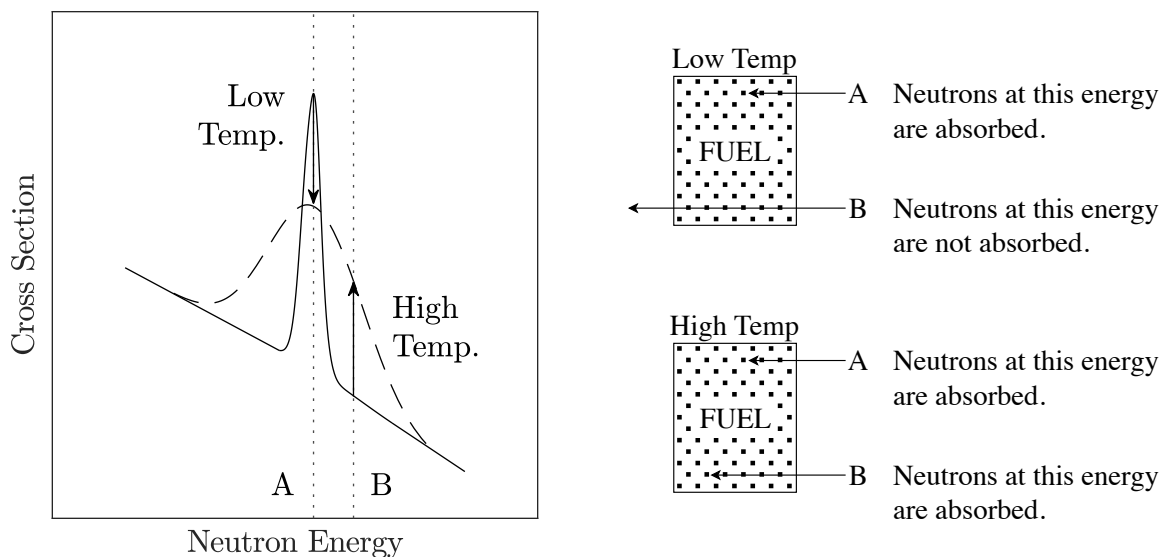


Figure 2.9: Self shielding of resonance peaks in UO_2 .

is that neutrons at energy B, and all other energies that were previously “off-resonance”, have an increased probability of absorption in the broadened resonance peak. As shown in Figure 2.9 neutrons at energy B escaped the fuel region without resonance capture at the low temperature, but are resonantly captured at the higher temperature. The fuel temperature rise, with its accompanying doppler-broadened resonance peaks, thus causes increased resonance absorption of neutrons above and below resonance energies with minimal reduction of neutrons absorbed at the resonance energy, which is a negative reactivity effect.

The self-shielding effect is observed in the heterogeneous system when the mean free path of the neutron is much shorter than the path length of the neutron in the absorbing medium. As the fuel temperatures increase, the mean free path of the neutrons approaches the fuel dimensions. The result is that the neutrons at the centroid energy of the resonance will experience an increased probability of passing through the fuel without capture. Now the increased capture rate of “off resonance” neutrons is accompanied by a decreased capture rate of neutrons at the resonance energy. The effect is that the capture rate is increased less for an increment of temperature rise starting from high temperatures (say increasing fuel temp from 2000°F than for the same incremental increase from an initially low temperature. (70°F to 71°F). The smaller change in neutron capture rate at high temperatures translates to a temperature coefficient of reactivity that decreases in magnitude with increasing temperature.

The fuel temperature coefficient changes as other reactor parameters change. Some of these changes are described below:

1. *Fuel temperature* – As fuel temperature increases, the effect of doppler broadening on reactivity per degree change in fuel temperature decreases because there is less increase in resonance capture per degree of fuel temperature rise. Therefore, the fuel temperature coefficient becomes less negative as fuel temperature increases.

2. *Power level* – As power level increases, the fuel temperature increases. As stated previously, this causes the fuel temperature coefficient to become less negative as power level increases.
3. *Core age* – As core age increases, the concentration of ^{240}Pu increases. ^{240}Pu has a larger resonance cross section than does ^{238}U . This causes the fuel temperature coefficient to become more negative over core life.

2.7.3 Isothermal Temperature Coefficient

The isothermal temperature coefficient is the combined effect of the moderator and fuel temperature coefficients due to changing operating temperature of the system as a whole. Consider, for instance, the condition where the reactor system is in thermal equilibrium and is either shutdown or at a power below the point of adding heat. This is referred to as an isothermal condition because the fuel, moderator, and reactor structural components are all at approximately the same temperature. A change in temperature affects both the fuel and the moderator, and the resulting reactivity effect is due to both the fuel and moderator temperature coefficients.

In practice, a change in moderator temperature will almost always produce a corresponding change in fuel temperature, even when the reactor is operating at power. For that reason, the published value of the moderator temperature coefficient is technically the isothermal temperature coefficient. This enables accurate calculations without increasing the complexity of the problem. This convention is utilized at the PULSTAR Reactor facility.

2.7.4 Void Coefficient

The PULSTAR does not have voids as a result of nucleate boiling at high power levels, but voids can be introduced near the core in the form of experiments. The change in reactivity per percent change in void concentration is called the void coefficient:

$$\alpha_{\text{void}} = \frac{\Delta\rho}{\Delta\%\text{Voids}} \quad (2.12)$$

A void adds reactivity to the reactor in much the same way as moderator temperature does. An increase in void concentration will cause a decrease in moderator density resulting in an increased slowing down length, increased diffusion length, and decreased neutron absorption in the moderator.

2.7.5 Power Coefficient

As reactor power increases there may be a change in average moderator temperature, fuel temperature, and void concentration. The combined effect is called the power coefficient. The power coefficient is defined as the change in reactivity per % change in reactor power level.

$$\alpha_{\text{power}} = \frac{\Delta\rho}{\Delta\%\text{Power}} \quad (2.13)$$

The power coefficient for the PULSTAR can be found in Reference 6.

2.7.6 Power Defect

The total reactivity effect of the power coefficient over the power range of the reactor is a more useful way of considering the effects of the power coefficient. This total reactivity effect is called the power defect:

$$\text{Power Defect} = (\alpha_{\text{power}})(\Delta\% \text{Power}) \quad (2.14)$$

In terms of the individual components of the Power Defect:

$$\text{Power Defect} = \alpha_{T,\text{mod}}\Delta T_{\text{avg}} + \alpha_{T,\text{fuel}}T_{\text{fuel}} \quad (2.15)$$

The power defect increases at a decreasing rate from the point of adding heat (POAH) to full power. The POAH is the power level where the fuel temperature begins to rise above the moderator temperature and is typically on the order of 1% full power.

The power defect changes over core life as a result of the changes in the moderator, fuel and void coefficients of reactivity. Although the fuel temperature contribution is responsible for most of the power defect, the change in the power defect does not necessarily follow the change in the fuel temperature coefficient over core life. The heat transfer properties of the fuel, pellet-to-clad gap, and fuel cladding determine the magnitude of the change in fuel temperature as power increases. As the core ages, the fuel cladding tends to creep in on the fuel causing better heat transfer. Therefore, a given power change at the end of core life results in a smaller change in fuel temperature with less broadening of the resonance peaks than at the beginning of core life. This tends to make the fuel temperature contribution to the power defect less negative over core life. Other factors affecting heat transfer from the fuel to the cladding, including fuel pellet swelling and contamination of the fill gas in the fuel pins, can cause the fuel temperature at the end of life to be higher or lower relative to the temperature at the same power early in core life. Since the build-up of ^{240}Pu makes the fuel reactivity effect at a given temperature larger at the end of core life, the power defect can either become more or less negative over core life. If the fuel temperature is the same or higher at the end of life, the power defect will be larger. However, the power defect can be smaller at the end of core life if the fuel temperature at a given power decreases and the decrease is enough to overcome the effect of the increase in the fuel temperature coefficient.

2.8 Reactivity Balances

Recall that reactivity, ρ , is defined as the fractional change in neutron population per generation. When the net reactivity is zero, the reactor is critical, and the power level remains constant with time. If ρ is positive, the reactor is supercritical, and power will increase with time. On the other hand, if ρ is negative, the reactor is subcritical and power level will decrease with time.

2.9 Components of Net Reactivity

The magnitude and sign of the reactor's reactivity depends upon several specific factors. These factors are either available to or can be calculated by the operator, as discussed in

the following sections. Most of this material has been covered in detail in the discussion of reactivity coefficients and will be reviewed only briefly. A detailed discussion of the effects of Xenon and Samarium is included.

2.9.1 Fuel

The excess reactivity of the core is used as the basis for reactivity balances. Excess reactivity is defined as the amount of reactivity in the core above that which is needed to go critical. For the PULSTAR, excess reactivity is referenced to cold, Xenon-free. The value of excess reactivity is calculated knowing specific reactor conditions and using Reference 6. Excess reactivity is dependent upon the time in core life (burn up). Excess reactivity is calculated using critical rods height is always positive.

2.9.2 Control Rods

The reactivity status of the core is influenced by the control rod position. The reactivity effect of control rods is negative. The value of this effect can be obtained from integral and differential rod worth curves. Note that the cumulative effect of all control rods in a bank is not equal to the simple algebraic sum of the individual control rod worths due to interaction of the high-worth control rod blades, and the value of control rod worth changes with burn up. Therefore, only the most recent and approved curves should be used.

2.9.3 Fission Product Poisons

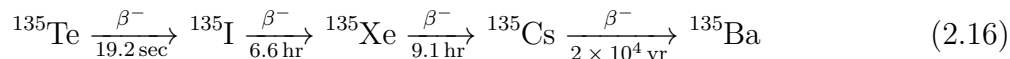
The fission process produces many daughter products, many of which are radioactive. These daughter products undergo an average of 3 decays before reaching a stable state, resulting in over 200 different fission products in a reactor core during operation.

Fission products with significant yields and high absorption cross-sections are called fission product poisons. The most important fission product poisons are Xenon and Samarium. The behavior of Xenon and Samarium during reactor operation should be well understood by reactor operators, because changing the concentrations of these poisons affects the reactivity of the core.

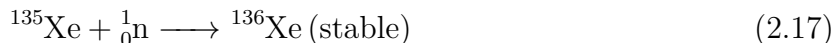
2.9.3.1 Xenon

Xenon is produced by decay of ^{135}I and as a fission product. At equilibrium, 97% of the Xenon is produced from ^{135}I decay, which is itself a product of radioactive decay of ^{135}Te . Because the half-life of ^{135}Te is much shorter than that of ^{135}I , it is reasonable to assume, for the purposes of our discussion, that iodine is produced directly from fission.

The decay scheme is as follows:



Xenon is removed by radioactive decay, as shown above, and by neutron capture or burnout. The burnout reaction is as follows:



^{136}Xe has a small absorption cross-section for neutrons, so the poison effect of the Xenon is effectively removed by burnout. Burnout accounts for about 67% of Xenon removal at the neutron flux levels encountered in a commercial reactor. Since the PULSTAR operates at much lower flux levels than a commercial plant, burnout is a very small part of the Xe removal process.

As illustrated by Figure 2.10, Xenon concentration is dependent on the reactor's operating history. Following a period of reactor operation at a constant power, an equilibrium condition is reached in which the rate of production of Xenon equals the rate of removal of Xenon. The concentration of both Xenon and iodine remain constant as long as power remains constant. The equilibrium value of Xenon reactivity is flux dependent; that is, the Xenon concentration at equilibrium is higher at 100% power than it is at 50% power.

When power is increased, Xenon burnout will increase while initially the production and removal decay processes remain essentially unchanged. The increase in production of Xenon from fission is insignificant (approximately 5% of total production). The net result is a reduction in Xenon concentration. As iodine concentration builds up, the production of Xenon increases, counteracting the increase in burnout. Xenon concentration reaches a minimum value 2-6 hours after the power increase and begins to increase. Approximately 40 hours after the power increase, Xenon concentration will reach a new, higher equilibrium value.

The value of Xenon reactivity at equilibrium is not a linear function of power. The equilibrium reactivity at 50% power is 80% of the equilibrium reactivity at 100% power. A typical PULSTAR value for equilibrium reactivity at 100% power is 900 pcm.

A reduction in power reduces burnout. Since the production from iodine decay is initially unchanged, Xenon concentration will increase. As N_{Xe} increases, the removal terms (decay and burnout) both increase. The production from decay of iodine decreases because iodine is decaying toward a lower equilibrium concentration proportional to power. These two effects cause Xe concentration to peak after 2-6 hours and then begin to decrease. After about 40 hours, Xenon equilibrium will be established at a level determined by the lower power.

When the reactor is shutdown from power operations, the burnout term is negligible. Production of iodine from fission is also negligible. Since ^{135}I has a shorter half-life than ^{135}Xe and the equilibrium concentration of ^{135}I is greater than the equilibrium concentration of ^{135}Xe , the production of Xenon by iodine decay will exceed removal by Xenon decay and Xenon concentration will increase. As before, a peak will occur when production equals removal. The time to peak is related to the operating history. The time to peak following shutdown is approximately the square root of the pre-shutdown power level. After the peak, Xenon concentration will decrease to a Xenon-free condition in approximately 70 hours. As

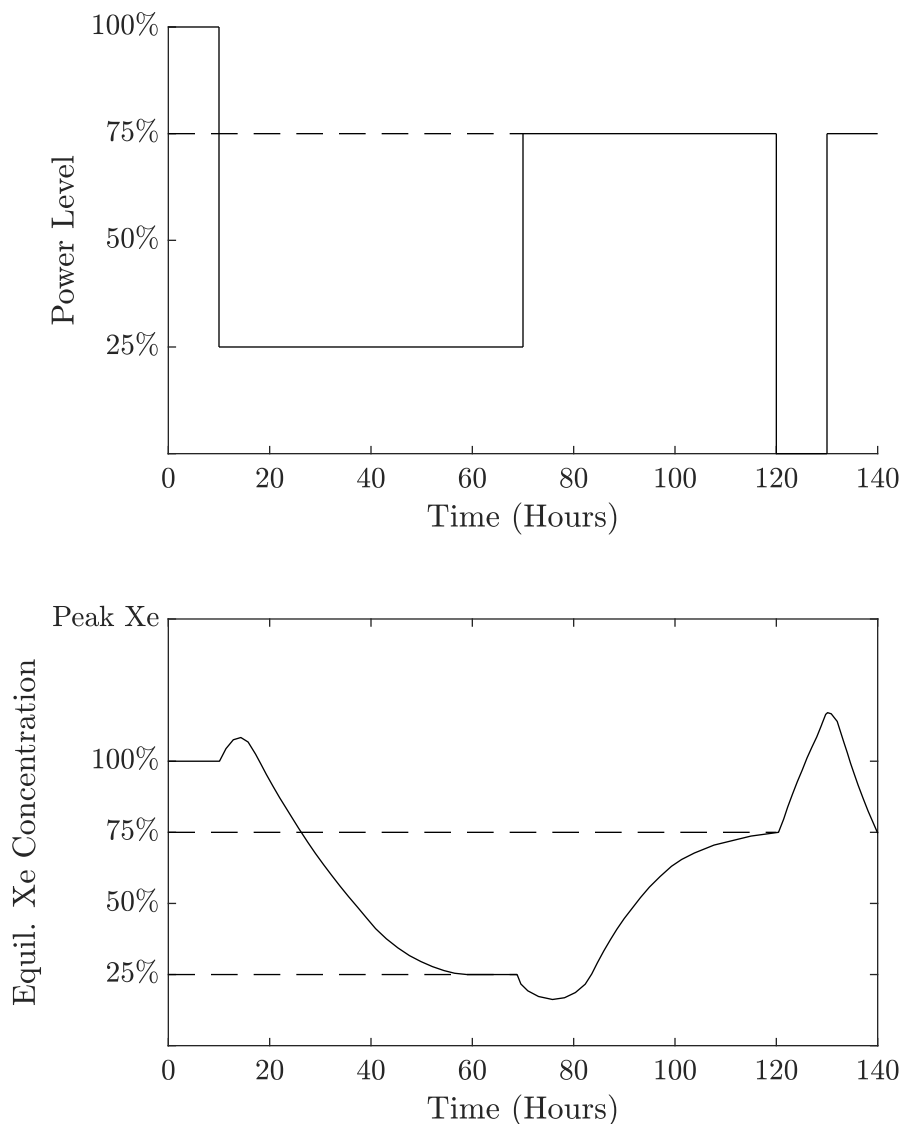


Figure 2.10: The effect of power changes on Xe concentration.

illustrated by Figure 2.11 the value of Xenon reactivity at the peak following a reactor shutdown is higher than the value at the equilibrium power prior to shutdown. For example, the peak reactivity after shutdown from 100% power is typically 1060 pcm.

2.9.3.2 Samarium

Samarium (Sm) is not as big a concern as Xenon because it has a smaller total reactivity effect and because the reactivity changes during reactor operation are relatively small.

Samarium is produced in the decay chain of Neodymium (Nd), which has a fission yield of 1.4%. The decay scheme is as follows:



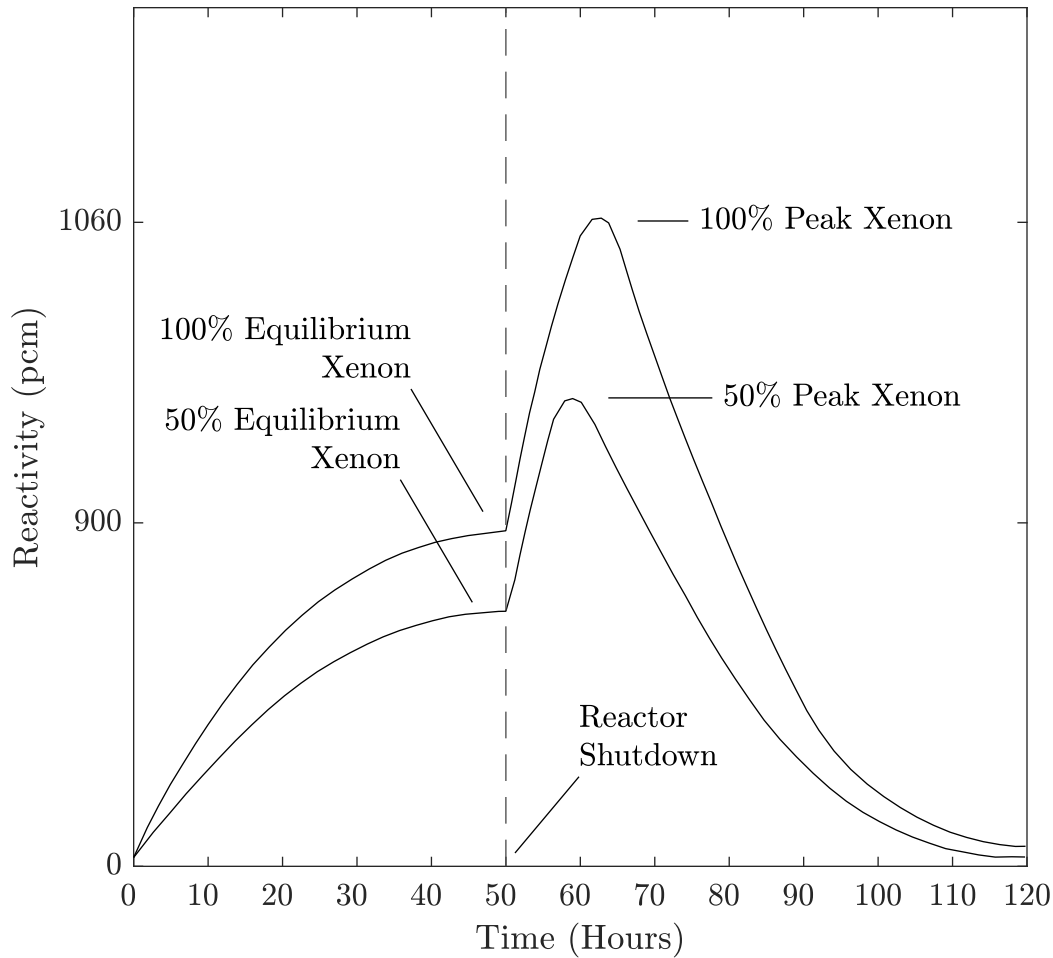
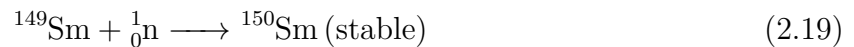


Figure 2.11: Peak Xe after shutdown.

We will consider the production of Promethium (Pm) to be directly from fission because of the comparatively short half-life of ^{149}Nd .

As shown in Equation 2.18, ^{149}Sm is essentially stable. Samarium is removed by neutron capture, or burnout, as follows:



Both the production and removal terms for Samarium are directly proportional to neutron flux, but the Samarium equilibrium concentration is independent of neutron flux. This is apparent when one realizes that a change in flux will change production and removal by the same amount. The effect of the flux change on equilibrium concentration, then, is canceled by these equal and opposite effects.

Once Samarium has built-up to equilibrium, which occurs in the first 40-50 days of reactor operation, it will remain there because of its long half-life. The reactivity added by the increase to equilibrium Samarium concentration is usually included in the excess reactivity calculations. This results in a reduction in core excess reactivity during the first 40-50 days

of operation.

The reactivity caused by the change in Samarium concentration from equilibrium must be included in reactivity balance calculations. When power changes, the change in burnout will immediately affect the Samarium concentration, but after 12-16 days, a peak (for down-power) or dip (for up-power) will occur. Equilibrium will be re-established, at the original concentration, in 40-50 days.

When the reactor is shutdown, ^{149}Pm decays to ^{149}Sm . Burnout is negligible and Samarium concentration increases to a peak value and stays there. The peak value is proportional to prior operating history. The increase in Samarium concentration after shutdown must be compensated for on the next startup. After startup, Samarium will burnout to the equilibrium value.

2.9.4 Moderator Temperature

Recall that a change in temperature of the moderator can add either positive or negative reactivity depending upon the direction of temperature change based on a reference temperature.

2.9.5 Power Defect

As power level is increased, the fuel temperature and average moderator temperature increase adding negative reactivity to the core. The value of the power defect at PULSTAR can be found in Reference 6.

2.9.6 Voids

At PWR's, voids are normally included as part of the power defect. The PULSTAR does not operate with nucleate boiling, so there is no void coefficient associated with heat production. There is, however, a coefficient that relates the size of a water displacement, or void, to reactivity. This is used to calculate the effect that a sample or experimental apparatus will have when placed near the core. The value of the PULSTAR void coefficient can be found in Reference 6.

2.9.7 Experiments

At a research reactor, such as the PULSTAR, there may be reactivity effects associated with experiments. The effects may be either positive or negative. Reference 6 lists the reactivity values for various experimental facilities. One effect that must be considered is the effect due to the startup source. Therefore, when doing reactivity balance calculations, if the source is out, it is worth -16 pcm and if the source is in, it is worth 0 pcm.

2.10 Reactivity Balances

By altering one or more of the aforementioned factors, we can produce a net positive, a net negative, or a net zero reactivity change. In the last case all the factors are in balance. We write a relationship that states the net reactivity as the sum of the reactivity components discussed in Sections 2.9 to 2.9.7.

$$\rho_{\text{net}} = \rho_{\text{excess}} + \rho_{\text{rods}} + \rho_{\text{temp}} + \rho_{\text{power}} + \rho_{\text{Xe}} + \rho_{\text{exp}} + \rho_{\text{source}} \quad (2.20)$$

This equation may be solved for any one component of ρ_{net} provided all the other reactivity components are known. In the case of an estimated critical position calculation, the equation is solved for ρ_{rods} and in the case of a shutdown margin calculation it is solved for ρ_{net} . These calculations will be performed later in this section.

In other cases, it may be necessary to solve the reactivity balance equation in terms of $\Delta\rho$. This is accomplished by subtracting Equation 2.20 at a known ρ_{net} (usually zero) and various individual components from another solution of Equation 2.20 with the same ρ_{net} and different individual components resulting in:

$$\Delta\rho_{\text{net}} = \Delta\rho_{\text{excess}} + \Delta\rho_{\text{rods}} + \Delta\rho_{\text{temp}} + \Delta\rho_{\text{power}} + \Delta\rho_{\text{Xe}} + \Delta\rho_{\text{exp}} + \Delta\rho_{\text{source}} \quad (2.21)$$

To understand the meaning of these relationships better, consider the following examples:

Example 2.3 :

A reactor is loaded with 2100 pcm ρ_{excess} the rods are partially withdrawn leaving -1800 pcm inserted in the core, there is -200 pcm Xenon in the core, and the temperature reactivity is -150 pcm. What is ρ_{net} ?

$$\begin{aligned} \rho_{\text{net}} &= (2100 - 1800 - 200 - 150) \text{ pcm} \\ &= -50 \text{ pcm} \end{aligned}$$

Example 2.4 :

The reactor is critical with -200 pcm Xenon and a -150 pcm temperature reactivity. After a shutdown the Xenon reactivity is -300 pcm and temperature reactivity is -100 pcm. How much reactivity must the rods add/subtract to take the reactor critical?

$$\begin{aligned} \Delta\rho_{\text{net}} &= \Delta\rho_{\text{excess}} + \Delta\rho_{\text{rods}} + \Delta\rho_{\text{temp}} + \Delta\rho_{\text{power}} + \Delta\rho_{\text{Xe}} + \Delta\rho_{\text{exp}} + \Delta\rho_{\text{source}} \\ \rho_{\text{rods}} &= -(\rho_{\text{temp}} + \rho_{\text{Xe}}) \\ &= -[(-100 - (-150)) + (-300 - (-200))] \\ &= -(50 - 100) \\ &= 50 \text{ pcm} \end{aligned}$$

2.11 Estimated Critical Position (ECP)

Prior to beginning a routine reactor startup, an estimate of the rod configuration at which the reactor will become critical is made. This estimate, normally known as an estimated critical position, is necessary to allow the operator to detect any unusual effects or malfunctions of the reactor system.

This estimate also allows the operator to predict normal reactor response to any reactivity adjustment he/she makes as he/she is approaching criticality. The degree of correspondence between predicted and actual critical position provides engineering with important data concerning core performance.

An ECP is derived by accounting for the various reactivity effects on the reactor prior to startup, and hence is a *zero-power reactivity balance*. These effects on the reactor and an estimate of the critical position can be determined by two methods:

1. Algebraically sum all the individual components of reactivity to the point where the net reactivity is zero. This is called the *total reactivity balance method*.
2. Balance all changes in the individual components of total reactor reactivity since the last known actual critical position (ACP). This is called the *incremental change method*.

At the PULSTAR, an ECP is calculated prior to the initial reactor startup of the day, to enable the operator to account for all reactivity effects. This estimate is compared to the actual critical position before 100 watts.

Ideally, the operator's ECP should be the same as the ACP. However, due to inaccuracies in reading graphs, Xenon concentration changes, and actual parameters being different from predicted values, it is possible that the ECP and ACP will not match exactly. To account for these effects and to remain within a realm of safety, a limit is placed on the ECP (± 200 pcm for PULSTAR). If the reactor fails to reach criticality by the time the upper limit is reached, or if the reactor goes critical prior to the lower limit, then the reactivity effects of the reactor are substantially different from what the operator expected. At this point, operations should stop and an evaluation of the situation should be made. This evaluation should include another computation of the ECP with particular attention paid to possible math errors and potential problems with the reactivity control system.

2.11.1 Incremental Change Method

The incremental change method of an ECP calculation is a core reactivity balance against the last known actual critical position (ACP). For a given set of conditions and through the use of various core reactivity graphs, the actual rod positions at which the reactor should go critical can be estimated using Equation 2.22.

$$\Delta\rho_{\text{ECP-ACP}} = \Delta\rho_{\text{rods}} + \Delta\rho_{\text{temp}} + \Delta\rho_{\text{power}} + \Delta\rho_{\text{Xe}} + \Delta\rho_{\text{exp}} + \Delta\rho_{\text{source}} \quad (2.22)$$

Note that $\Delta\rho_{\text{fuel}}$ ($\Delta\rho_{\text{excess}}$) is assumed to equal zero and is not included in the reactivity balance. Note, also, that the difference between the sum of the individual reactivity components between the ECP and ACP is the same as the sum of the difference of the individual reactivity components between the ECP and ACP. This is illustrated by the following example.

Example 2.5 :

At the last ACP, Xenon reactivity was -200 pcm, temperature reactivity was -100 pcm, and rod reactivity was -800 pcm. At the ECP, Xenon reactivity is -250 pcm and temperature reactivity is -50. What is rod reactivity at the ECP?

1. Using sum of differences (ECP - ACP)

$$\begin{aligned}\Delta\rho_{\text{rods}} &= -(\Delta\rho_{\text{temp}} + \Delta\rho_{\text{Xe}}) \\ &= -[(-50 - (-100)) + (-250 - (-200))] \\ &= -(50 - 50) \\ &= 0 \text{ pcm}\end{aligned}$$

2. Using difference of sums (ECP - ACP)

$$\begin{aligned}\rho_{\text{ECP}} &= -(\rho_{\text{temp}} + \rho_{\text{Xe}}) \\ &= -50 - 250 \\ &= -300 \text{ pcm} \\ \rho_{\text{ACP}} &= -(\rho_{\text{temp}} + \rho_{\text{Xe}}) \\ &= -100 - 200 \\ &= -300 \text{ pcm} \\ \Delta\rho_{\text{rods}} &= -(300 + (-300)) \\ &= 0 \text{ pcm}\end{aligned}$$

Table 2.1 shows an estimated critical position calculation using the incremental change method. Note that only the components of net reactivity that change are included.

Example 2.6 :

The following data applies to the calculation in Table 2.1:

- | | | | |
|----------------|-----------------|----------------|---------|
| 1. ACP at 10 W | = 18.8" on gang | 2. ECP at 10 W | = ? |
| Mod. Temp. | = 99.5 °F | Mod. Temp. | = 90 °F |
| Source | = In | Source | = In |

Table 2.1: Incremental change reactivity balance.

Reactivity Component	$\rho_{\text{final}}(\text{ECP})$	$\rho_{\text{initial}}(\text{ACP})$	ρ_{net} (Final-Initial)
Power	0	0	0
Xenon	-213 pcm	-68 pcm	-145 pcm
Source	0	0	0
Mod. Temp.	$(90 - 99.5) \times -3.9 \text{ pcm}/^\circ\text{F}$		+37 pcm
	SUM >		-108 pcm

$$\rho_{\text{rods}}(\text{ACP}) = -910 \text{ pcm}$$

$$\Delta\rho_{\text{rods}} = +108 \text{ pcm}$$

$$\begin{aligned}\rho_{\text{rods}}(\text{ECP}) &= \Delta\rho_{\text{net}} \\ &= -802 \text{ pcm}\end{aligned}$$

New critical rod height corresponds to -802 pcm rod reactivity inserted in core, $\approx 19.2''$ gang.

2.11.2 Total Reactivity Balance Method

The total reactivity balance method is a sum of all the component reactivities that exist at the time of the ECP. By knowing all the individual components, the reactivity of the rods necessary to make the net reactivity equal to zero (reactor critical) can be determined. The individual components can be calculated by knowing certain reactor parameters or can be read off of curves. The total reactivity balance method involves solving:

$$\rho_{\text{rods}} = -(\rho_{\text{excess}} + \rho_{\text{temp}} + \rho_{\text{power}} + \rho_{\text{boron}} + \rho_{\text{Xe}} + \rho_{\text{exp}} + \rho_{\text{source}}) \quad (2.23)$$

Example 2.7:

Calculate the ECP using the following data and Figure 2.12.

$$\begin{aligned}\text{Cold Clean Excess} &= 967 \text{ pcm} \\ \text{Temperature Reactivity} &= -100 \text{ pcm} \\ \text{Xenon Reactivity} &= -200 \text{ pcm} \\ \text{Source} &= \text{In}\end{aligned}$$

$$\rho_{\text{rods}} = -((967) + (-100) + (-200)) = -667 \text{ pcm}$$

Then using -667 pcm as the control rod reactivity inserted in the core, the ECP can be determined from the integral rod worth curves.

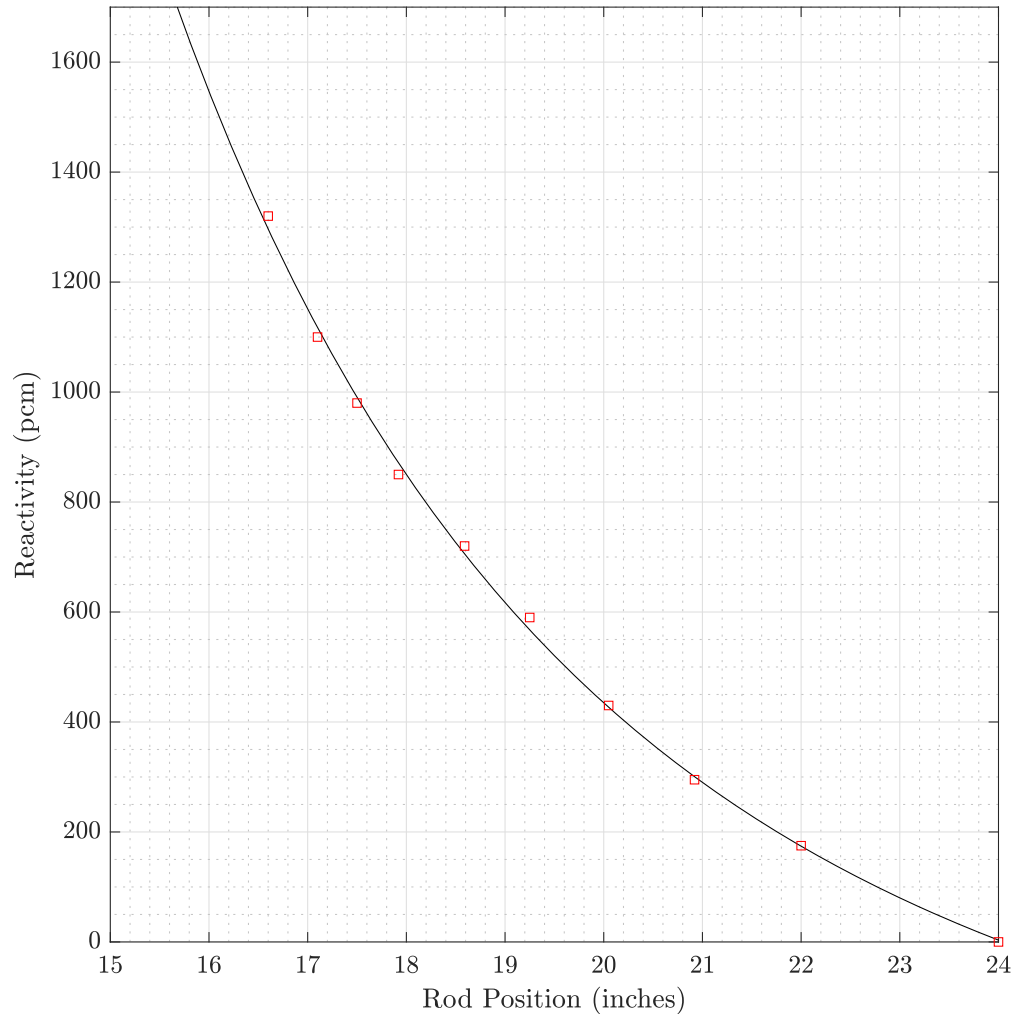


Figure 2.12: PULSTAR ganged integral rod worth.

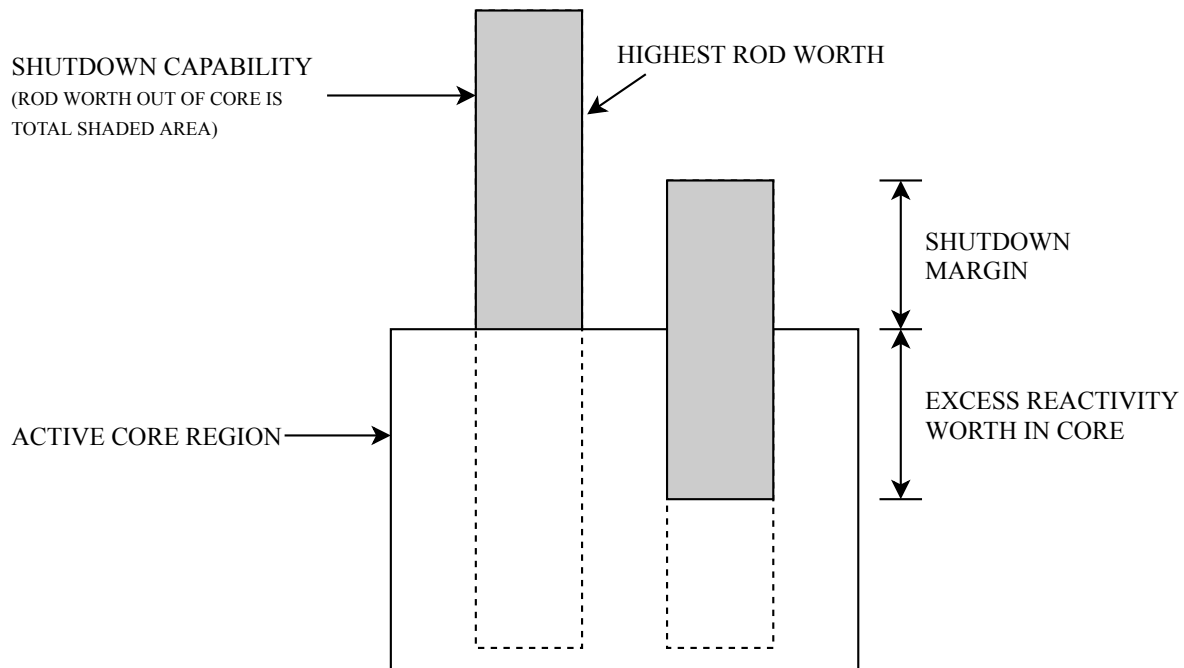
2.12 Shutdown Margin

A reactivity balance can also be used to calculate Shutdown Margin (SDM). SDM is defined as the margin by which the reactor is (or can be) shutdown. In its general use, shutdown margin is simply $(1 - K_{\text{eff}})$, but when applied to technical specification requirements it is the amount by which the reactor can be shutdown from a given condition.

At the PULSTAR, the shutdown margin limit is the “minimum shutdown reactivity necessary to provide confidence that the reactor can be made subcritical by means of the control and safety systems, starting from any permissible operating condition, and that the reactor will remain subcritical, without further operator actions.”

This implies that the ability to shutdown the reactor and to keep it shutdown exists even if the highest worth control rod is fully withdrawn, the temperature decreases, and Xenon decays, etc. The shutdown margin calculation is a verification that the limit is met in a cold, clean condition. The definition and value of the minimum SDM are always found

in the technical specifications for a facility. The minimum value of the SDM is 400 pcm at PULSTAR. Figure 2.13 shows a pictorial representation of shutdown margin and excess reactivity.



$$\text{SHUTDOWN MARGIN} = \text{TOTAL ROD WORTH} - \text{EXCESS REACTIVITY} - \text{HIGHEST WORTH ROD}$$

Figure 2.13: Shutdown margin.

2.13 Other Reactivity Balances

Besides ECP and SDM calculations, other types of reactivity balances may be performed. They include: (1) prediction of new rod position for changes in power level and/or temperature, (2) determination of reactivity anomalies (i.e. net calculated reactivity does not equal zero when critical), and (3) changes in Xenon and Samarium concentrations.

When the reactor is exactly critical, $\rho_{\text{net}} = 0$. This is true regardless of plant conditions. With the known conditions of power, temperature, Xenon concentration, etc., a reactivity balance for ρ_{net} will give a result of 0 when the reactor is critical.

An example of a reactivity balance to determine rod position for a given SUR and peak power reached follows:

Example 2.8:

The PULSTAR is critical at 10 watts, ACP. = 20.0" gang. A 1.5 dpm SUR is desired. What is the rod position required to obtain this SUR? If rods are moved to that position, and reactor power is allowed to increase with no further rod motion, what is the peak power reached by the reactor?

Solution: First determine the reactivity required for a 1.5 dpm SUR. Therefore, to get a 1.5 dpm SUR, 267 pcm above criticality must be added. Second, determine the supercritical rod height for the 1.5 dpm SUR by looking at Figure 2.12. To add 267 pcm, rods must move from 20.0" to 21.3".

$$\begin{aligned}\rho &= \frac{\text{SUR} \times \beta_{\text{eff}}}{26.1 \times \lambda_{\text{eff}} + \text{SUR}} \\ &= \frac{1.5 \times 730}{26.1 \times 0.1 + 1.5} \\ &= 267 \text{ pcm}\end{aligned}$$

The first question has now been answered. Next, we must determine the peak power attained by the reactor. Recall that the question specifies no further rod motion after the 1.5 dpm SUR is established. This means that any subsequent reactivity changes will be due to reactivity feedback from the reactor fuel and the primary coolant.

As long as reactor power is below the point of adding heat (POAH) there is no reactivity feedback and the SUR is constant. However, as soon as power exceeds the POAH, the fuel temperature begins to increase and doppler adds negative reactivity. After some delay, the heat produced in the fuel is transferred to the moderator and the moderator temperature coefficient adds reactivity.

As long as $\rho_{\text{net}} > 0$, there will be a positive SUR and reactor power will continue to increase. Once this fact is realized, it is obvious that the peak power is reached when $\rho_{\text{net}} = 0$ and the reactor is exactly critical. Therefore, use a total reactivity balance to determine the change in reactivity due to power.

$$\begin{aligned}\Delta\rho_{\text{net}} &= \Delta\rho_{\text{rods}} + \Delta\rho_{\text{power}} = 0 \\ \Delta\rho_{\text{power}} &= -\Delta\rho_{\text{rods}} \\ &= -267 \text{ pcm}\end{aligned}$$

Now using the Power Coefficient, we can determine the resulting change in power:

$$\begin{aligned}\rho_{\text{power}} &= \alpha_{\text{power}} \times \Delta\text{Power} \\ \Delta\text{Power} &= \frac{\rho_{\text{power}}}{\alpha_{\text{power}}} \\ &= \frac{-267 \text{ pcm}}{-330 \text{ pcm/MW}} \\ &= 0.809 \text{ MW}\end{aligned}$$

2.14 Questions

- Given that reactor power increased from 100 watts to 1.5 kW in 2 minutes, determine: the startup rate, doubling time, and decade time. Also, determine the reactivity addition causing this power increase. List all assumptions.
- Given that at the beginning of life (BOL) the effective delayed neutron fraction, β_{eff} , for the core is 700 pcm and at end of life (EOL) β_{eff} is 540 pcm, calculate the reactivity required at BOL and EOL for a 1.0 DPM SUR. List all assumptions. How does this affect reactor control?
- Suppose the reactor is being maintained in a shutdown condition with $\beta_{\text{eff}} = 0.99$. Assume that the temperature coefficient of reactivity is $-3.9 \frac{\text{pcm}}{^\circ\text{F}}$. Calculate the resultant startup rate that would be generated by cooling the pool down from 110 °F to 101 °F. List all assumptions.
- If a rod withdrawal of +300 pcm increased reactor power by 20%, what is the power coefficient?
- The reactor is critical in the intermediate range with $T_{\text{avg}} = 105^\circ\text{F}$. If filling a beam tube adds +50 pcm of positive reactivity, what is the final T_{avg} ?
- Given: Reactor at 1.0 MW, ρ_{rods} (in core) = -660 pcm (19"), Average Moderator Temperature = 100 °F, Xenon reactivity = -100 pcm, neutron source out. Calculate: Cold, clean, excess reactivity. Use PULSTAR constants and the reference conditions given below. Reference conditions for ρ_{excess} :

T_2	= 70 °F
Power	= 10 W
Xenon	= 0
Samples	= None
Source	= In
- Using the excess reactivity from Question 6, calculate the ECP for the following conditions:

Xenon reactivity	= -122 pcm
T_{avg}	= 70 °F
Samples	= None

Show work. Use PULSTAR constants.
- The reactor is at 10 watts with all rods at 19.0".
 - Calculate the required Reg Rod position to give a 1.5 DPM SUR if the Reg Rod differential worth is 300 pcm per inch between 18.0" and 19.0" Use the one-group approximation to the Inhour equation to calculate reactivity.
 - Sketch power vs. time for this transient, assuming no change in Xenon and no secondary heat load.

9. Draw a graph of Xenon reactivity vs. time for the following power history:
- Time = 0 hours Startup to 50% power from Xenon free.
 - Time = 40 hours Increase power to 100%
 - Time = 90 hours Decrease power to 80%
 - Time = 150 hours Shutdown. (Show time to Xenon free.)

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- [6] *PULSTAR Data Summary*, volume 2.

3 Reactor Operations

A clear understanding of the neutron flux profile and the reactions rates present in an operating nuclear reactor is essential to knowledgeable operation of the PULSTAR reactor.

3.1 Neutron Flux

The neutron flux may be simplified by considering it to have two components. These are the radial or horizontal and the axial or vertical terms. In the radial direction, consideration is given to the fuel pellet, and coolant channels. The axial component of the thermal neutron flux is primarily a function of control rod position at the PULSTAR reactor.

The neutron flux in the fuel pellet is highest at the edges of the pellet and dips in the center due to self-shielding of the thermal neutron flux (Figure 3.1).

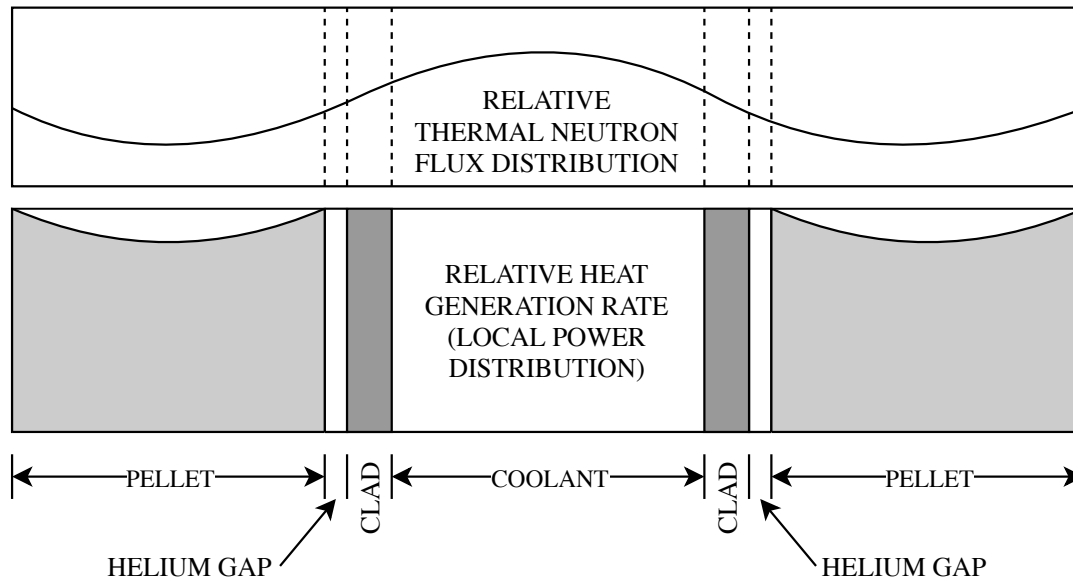


Figure 3.1: Local radial flux distribution.

The axial peak in the thermal neutron flux occurs near the mid-plane of the core and decreases at the top and bottom (Figure 3.2).

This is a simplified symmetrical illustration. Thermal neutron flux is subject to distortion by the control rods and fuel burn-up.

Several different terms are employed for describing unfavorable flux distribution conditions. As an example, consider these definitions listed in a typical Final Safety Analysis Report (FSAR).

1. *Linear Power Density*: the power generated in a given length of fuel rod, usually one foot. It is the integral of the heat flux divided by the heat transfer length, and is usually measured in kW/ft.

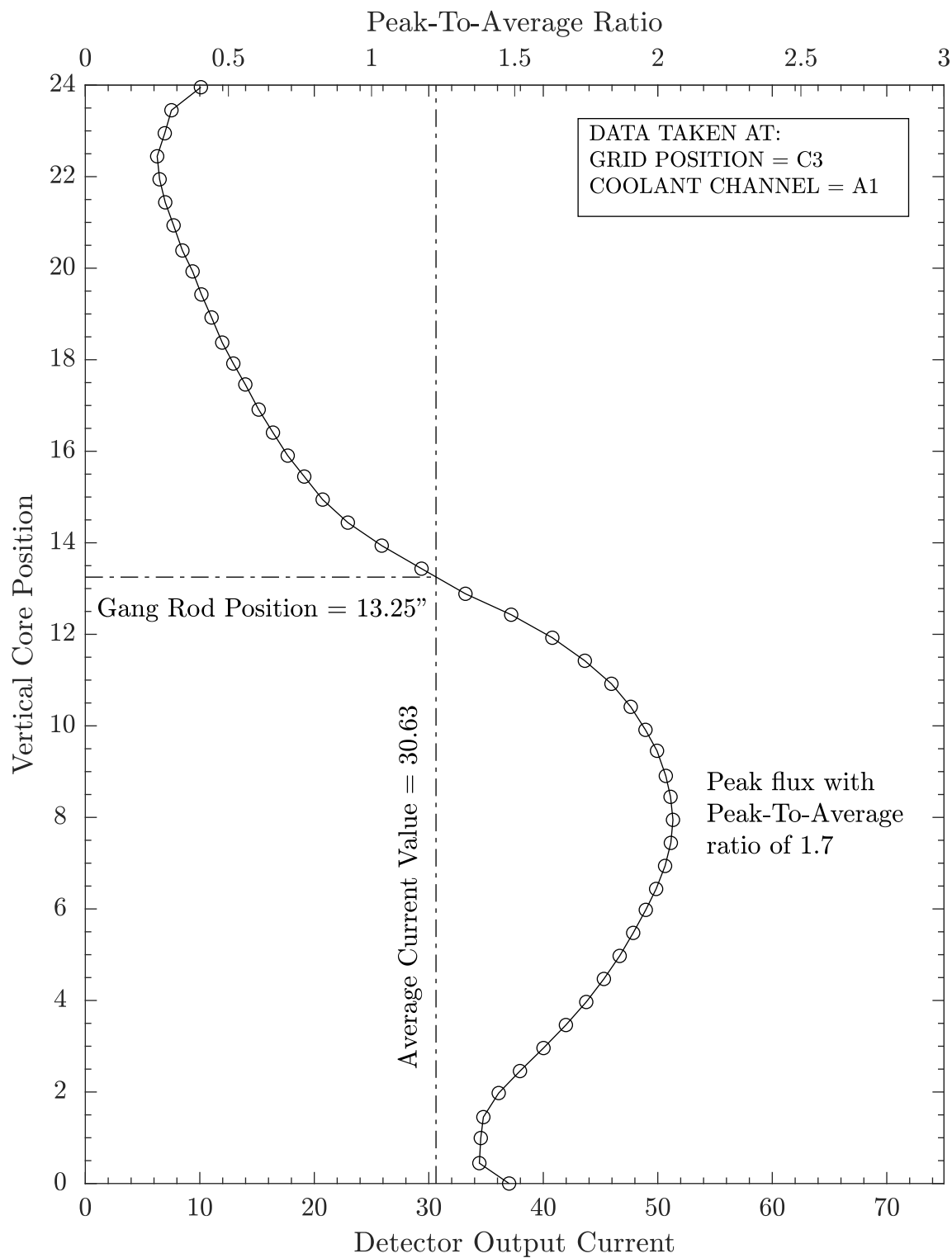


Figure 3.2: In-core axial flux distribution NCSU PULSTAR reactor.

2. *Radial Peak-to-Average Ratio (RPA)*: the ratio of the peak power density to the average power density in the radial (horizontal) direction.
3. *Axial Peak-to-Average Ratio (APA)*: the ratio of the peak power density to the average power density in the axial (vertical) direction.

3.2 Flux Distribution

The ideal flux distribution is uniform throughout the core, but zero at the core boundary. If these conditions were achievable, then all regions of the core would be generating the same amount of power. Actually, the thermal flux distribution is a series of local peaks and dips that depend on many variable factors. There are various techniques that are used in an attempt to idealize flux distribution.

A neutron reflector is one method of returning leakage neutrons to the core (Figure 3.3). The PULSTAR uses water and graphite as reflectors. Some of the fast and thermal leakage neutrons are scattered back into the core. Because of the moderating characteristics of the reflector, there is a peak in the thermal flux (reflector peak) which is higher than the thermal flux at the core edge. Because neutrons behave much like a gas and tend to flow or migrate from areas of high density to areas of low density. As a result, some of the reflected neutrons will migrate back into the core, thus flattening out the overall flux distribution.

Insertion of a control rod will depress the local flux causing a flux peak somewhere else in the core. This can have an effect on the indicated power on the NI system and the operator must be aware of this whenever he or she is making adjustments to Safety #1 and Safety #2 control rod height.

Core structural materials also have an effect on flux distribution. Areas where there is a large amount of moderation will have high local thermal fluxes. Spacer grids and other structures will have low local thermal fluxes. These effects can be minimized by good engineering design.

3.3 Power Density

Power distribution ($\frac{\text{kW}}{\text{ft}}$) and the heat flux ϕ_q ($\frac{\text{BTU}}{\text{hr ft}^2}$) are directly proportional to the thermal neutron flux; therefore, these parameters also reach peak values near the core mid-plane.

$$\text{Power Density} = C \Sigma_f \phi_{th} \quad (3.1)$$

where: C = constant in terms of watt·sec/fission, Σ_f = macroscopic cross-section for fission, ϕ_{th} = thermal neutron flux.

Looking at the above equation and the fact that flux distribution is not equal, it can be seen that power is not generated equally throughout the core. The power or heat produced must be removed from the core to prevent core damage. The PULSTAR FSAR analysis takes into account the uneven power distribution by using two terms, axial hotspot factor (1.51) and

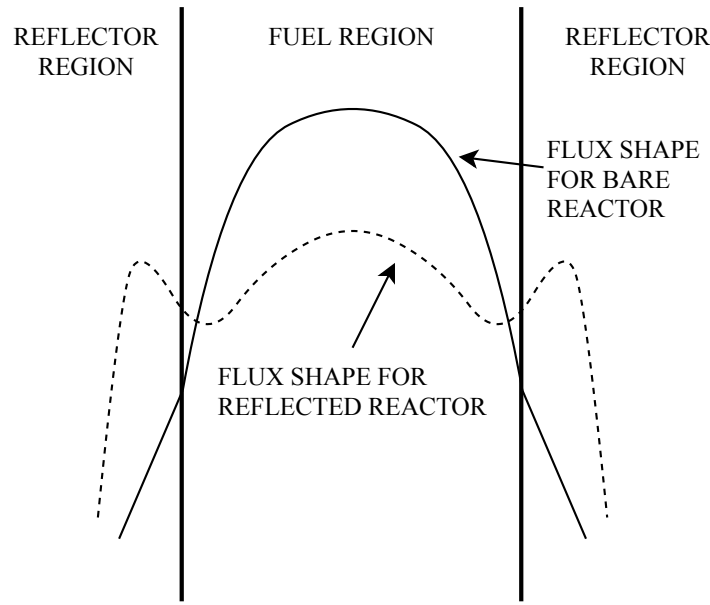


Figure 3.3: Effect of a reflector on flux shape.

radial hotspot factor (1.94) in the designed assumed worse case hotspot factor is 2.92.

It can be seen that most of the factors effecting flux distribution are of engineering nature and cannot be directly controlled by the operator. However, with a good basis for understanding the limits imposed on operation, the operator can intelligently and conscientiously ensure the safe operation of the reactor.

3.4 Power Range Maneuvers

In the power range, the operations required by the operator to level off at a constant power are different from the corresponding operations in the intermediate range. Correspondingly, the actions required to reduce power are also different.

A rod withdrawal from the 10 watts critical condition will place the reactor on a stable startup rate. As reactor power is increased above the point of adding heat (POAH), approximately 1% of full power, the fuel temperature begins increasing. The effect of the power coefficient inserts negative reactivity and decreases the rate of change of power. When the negative reactivity inserted by the power coefficient (the power defect) equals the positive reactivity inserted by the rods, the reactor will be critical at some power level in the power range.

If the control rods are withdrawn further, power will begin to increase again until enough negative reactivity is inserted by the power coefficient to compensate for the positive reactivity addition. A negative reactivity insertion (less than the power defect) in the power range will cause the reactor to go subcritical and reactor power will decrease until the positive reactivity inserted by the power coefficient brings the core's reactivity status back to zero.

Reactor power will level out at some lower power determined by the amount of reactivity inserted. For negative reactivity additions greater than the power-defect at the power that the reactor is operating, the reactor will go subcritical; power will decrease out of the power range, continue through the intermediate range, and level off at some equilibrium subcritical multiplication level in the source range. The reactivity feedback mechanism that is observed in the power range will be demonstrated in this experiment.

3.5 Natural Circulation

Natural circulation is the flow of fluid due to a density difference. For natural circulation to initiate, there must be (1) a heat source and (2) a heat sink, (3) the heat sink must be higher than the source, and (4) a complete path for flow must exist. Natural circulation is an important safety phenomenon that exists during loss of flow conditions in a reactor. Natural circulation may be demonstrated at the PULSTAR since its design allows operation of the reactor up to 100 kW with cooling by natural circulation in the pool. The core of the PULSTAR sits on a plenum. A large hole in the side of the plenum is covered by a disc called the flapper. When forced flow exists through the PULSTAR core, differential pressure, plenum-to-pool, developed by the primary pump holds the flapper shut. Upon loss of forced flow, the flapper hinges back, opening the plenum to provide a path for natural circulation. In this configuration: (1) the heat source is the reactor core, (2) the heat sink is the air at the surface of reactor pool, (3) the heat sink provide a path for natural circulation. In this configuration: (1) the heat source is the reactor core, (2) the heat sink is the air at the surface of reactor pool, (3) the heat sink is 20 feet above the top of the core, and (4) the continuous path for flow is into the plenum, through the core, into the pool, and back into the plenum. Natural circulation will be demonstrated by measurement of thermo-couple temperatures at the bottom and top of the core and by visual observation of thermal currents in the reactor pool.

3.6 Nominal Power Determination

To calibrate the power range nuclear instrumentation, a heat balance is normally performed to determine the nominal power at which the reactor is operating. This nominal power determination can then be used as the basis for calibrating the power range nuclear instrumentation - thus compensating for the effects of age and drift on the power range detectors and instruments and for long-term shifts in the neutron flux distribution. A neat balance is an energy balance, i.e., the sum of the heat sources at steady state must equal the sum of heat losses around the loop. The heat loss or gain across a device, such as a steam generator or heat exchanger, is equal to the product of the mass flow rate through the device, the heat capacity of the coolant, and the resulting temperature drop across the device.

$$\dot{Q} = \dot{m}c_p\Delta T \quad (3.2)$$

For the primary system in a PWR, the heat gains are from the reactor and pumps while the heat losses are from the steam generators and ambient losses. In the case of the PULSTAR's primary system, the only heat gain is from the reactor (the primary pump is only 7.5 HP)

and the heat losses are from the heat exchanger, ambient losses, and pool heat-up. The heat loss due to pool heat-up can be physically interpreted as the heat being taken out of the system to raise the bulk temperature of the pool.

At a research and training reactor like the PULSTAR, the most obvious method of determining thermal power output might be based upon the flow rate of the primary system and the temperature rise of the primary coolant from core inlet to core outlet. However, because of coolant mixing and the physical design of the PULSTAR Reactor, an accurate measurement of the coolant temperature entering the core is difficult to obtain. Therefore, the most accurate determination of reactor thermal power is to calculate a heat balance based upon primary coolant flow rate and the temperature drop across the primary side of the heat exchanger.

In any heat balance, it is essential to consider all sources of heat and all paths of heat loss. Ambient losses may occur in the form of heat-up of coolant pipes and/or the shielding and structural components. These extraneous heat losses can be minimized by establishing a steady state thermal condition in the system. Steady state implies that the system as a whole (coolant, piping, shielding, and structural components) remains at a constant temperature during the course of the heat balance.

If a true steady state thermal condition cannot be achieved, corrections for slow coolant heat-up or cool-down rates have to be made. In the same manner as the heat gain (or loss) from a device can be calculated using Equation 3.1 for a steady flow process, the heat gain (or loss) due to pool heat-up (or cool-down) can be calculated using: Equation 3.2 below for a uniform state process.

$$\dot{Q}_{\text{heat-up}} = MC_p \Delta \dot{T} \quad (3.3)$$

In the latter case the (M) represents the mass of the system (lbm) and $\Delta \dot{T}$ represents the heat-up or cool-down rate ($\frac{^\circ\text{F}}{\text{hr}}$). Note that the units are consistent between Equation 3.1 and Equation 3.2. If the pool or primary system is heating up, the $\Delta \dot{T}$ term will be positive; conversely, if a cool-down is occurring, $\Delta \dot{T}$ will be negative. An example will best illustrate the concept of a heat balance.

Example 3.1 :

The PULSTAR Reactor is operating with a heat-up rate of $1 \frac{^\circ\text{F}}{\text{hr}}$ and a ΔT across the primary side of the heat exchanger of 13.2°F . What is the nominal power of the reactor? Also given: Primary Flow = 500 gpm, Primary Volume = 15,650 gal, $c_p = 0.998 \frac{\text{BTU}}{\text{lbm}^\circ\text{F}}$, Density $\text{H}_2\text{O} = 8.272 \frac{\text{lbm}}{\text{gal}}$ and $1 \text{ MW} = 3.413 \times 10^6 \frac{\text{BTU}}{\text{hr}}$.

$$\begin{aligned} \text{Heat gains} &= \text{Heat losses} \\ \text{Reactor Power} &= \text{Heat exchanger losses} + \text{Pool heat-up} \\ &\quad + \text{Ambient losses (assumed equal to 0)} \end{aligned}$$

$$\begin{aligned}
 \dot{Q} &= \left(500 \frac{\text{gal}}{\text{min}}\right) \left(8.272 \frac{\text{lbm}}{\text{gal}}\right) \left(60 \frac{\text{min}}{\text{hr}}\right) \left(0.998 \frac{\text{BTU}}{\text{lbm} \cdot ^\circ\text{F}}\right) (13.2 ^\circ\text{F}) \\
 &+ (15\,650 \text{ gal}) \left(8.272 \frac{\text{lbm}}{\text{gal}}\right) \left(0.998 \frac{\text{BTU}}{\text{lbm} \cdot ^\circ\text{F}}\right) \left(1 \frac{^\circ\text{F}}{\text{hr}}\right) \\
 &= 3\,269\,161 \frac{\text{BTU}}{\text{hr}} + 129\,198 \frac{\text{BTU}}{\text{hr}} \\
 &= 3\,398\,359 \frac{\text{BTU}}{\text{hr}} \\
 &= 3\,398\,359 \frac{\text{BTU}}{\text{hr}} \left(\frac{1 \text{ MW}}{3.413 \times 10^6 \frac{\text{BTU}}{\text{hr}}}\right) \\
 &= 0.996 \text{ MW}
 \end{aligned}$$

It should be noted that besides the temperature-flow power determination, one other channel is installed at the PULSTAR that is unaffected by the flux tilts the N-16 Channel. This channel is therefore employed to verify that the nuclear instrumentation indication is within acceptable limits before an absolute heat balance power determination can be made.

3.7 Questions

1. What is the worst case Hot Spot Factor for the PULSTAR reactor?
2. Explain what effect a reflector has on flux shape.
3. The PULSTAR is operating at power and conditions are steady state. The primary side of the heat exchanger has a $\Delta T = 13.7 ^\circ\text{F}$ What is the actual power of the reactor?

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